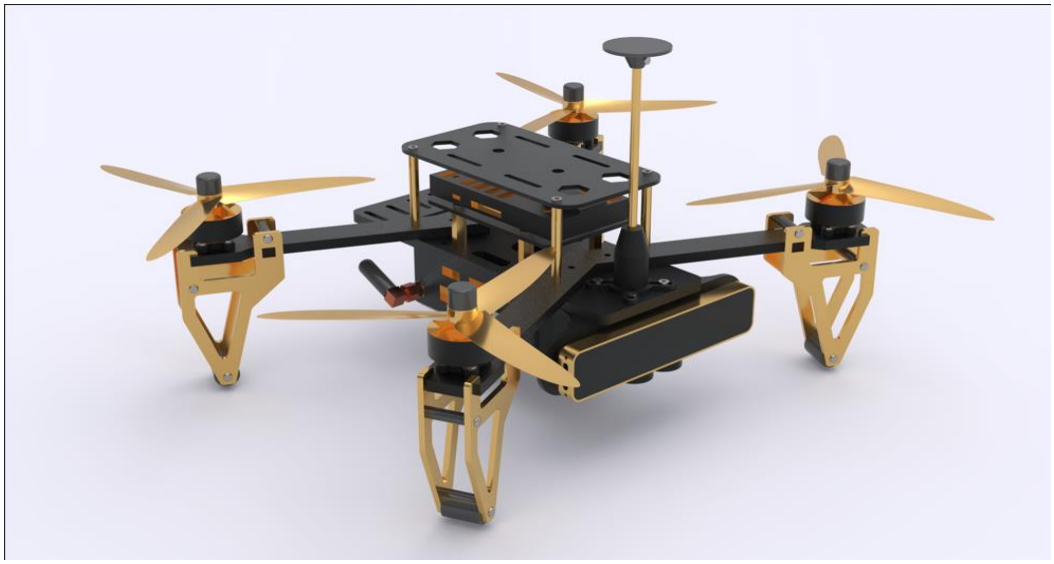
Lebanese American University

LAU School of Engineering

MEE351: Computer Aided Design

PX4 Drone



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What are px4 drones?

- PX drones refer to a family of drones made by PX4, which is an open-source autopilot software. These drones are used for various purposes, such as aerial photography, surveying, mapping, agricultural monitoring, search and rescue operations, and infrastructure inspection. They can be equipped with different sensors and cameras depending on the specific use case.

- PX drones fly using a combination of **rotors**, **motors**, and **control algorithms** to achieve stability and movement.
- **Lift and Thrust**: The drone is powered by four rotors (in a quadcopter configuration), each connected to a motor. The rotors spin at high speeds to generate lift, allowing the drone to ascend. The amount of lift produced depends on the speed at which the rotors spin.
- **Movement Control**:
    - **Pitch (Forward/Backward)**: By adjusting the speed of the front and rear rotors, the drone tilts and moves in the forward or backward direction.
    - **Roll (Side-to-Side)**: Adjusting the speed of the left and right rotors causes the drone to tilt and move side to side.
    - **Yaw (Rotation)**: The drone rotates around its vertical axis by varying the speeds of opposite rotors (front-back vs. left-right).
    - **Altitude (Up/Down)**: Increasing or decreasing the collective thrust of all four rotors causes the drone to ascend or descend.

Bill of Material:

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	main frame (v)		1
2	V bracket (V)		2
3	Part1 Battery Frame Spacer (K)		4
4	Part2 Battery Frame (K)		1
5	Screw battery frame (V)		4
6	Screw V Bracket main frame (V)		10
7	Part4 Companion Computer (K)		1
8	Part3 Companion Computer Antenna (K)		2
9	Part6 Flight controller frame spacer (K)		4
10	Part7 Flight Controller frame (K)		1
11	pixhawk 4 (V)		1
12	screw flight controller frame (V)		4
13	Screw core frame (V)		2
14	Part9 GPS base (K)		1
15	Part12 GPS screw cap (K)		1
16	Part10 GPS long rod (K)		1
17	Part11 GPS mount (K)		1
18	pin for rod upper (V)		1
19	screw GPS base (V)		4
20	Part13 GPS (K)		1
21	pin for GPS base (V)		1
22	servo motor F60PRO IV V2.0 KV1750 (V)		4
23	Propeller 6045R (V)		4
24	Part5 Servo Motor Covor (K)		4
25	Pin spacer thin (V)		8
26	pin spacer thick (V)		4
27	screw servo motor (V)		16
28	Part15 Landing leg spacer (thin) (K)		8
29	Part16 Landing Leg (K)		8
30	Part14 Landing leg spacer (thick) (K)		4
31	structure core frame (V)		1
32	Part17 Lidar Mount (K)		1
33	Part18 Lidar (K)		1
34	Part8 Frame T265 (K)		1
35	Screen T265(V)		1
36	Camera Assembeled		1
37	Platform		1

Figure 1:Bill of material (full assembly)

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Part19 Camera Sphere		1
2	Part20 Camera Base Arm		1
3	Camera Lens (V)		1
4	arm base (V)		1
5	Camera Arm (V)		1

Figure 2:Bill of material (assembly of Camera)

Parts with a (v) next to them were drawn by Vartan fakhreddine, while the ones with (K) were drawn by Khalid Kaddoura.

Rendered images:

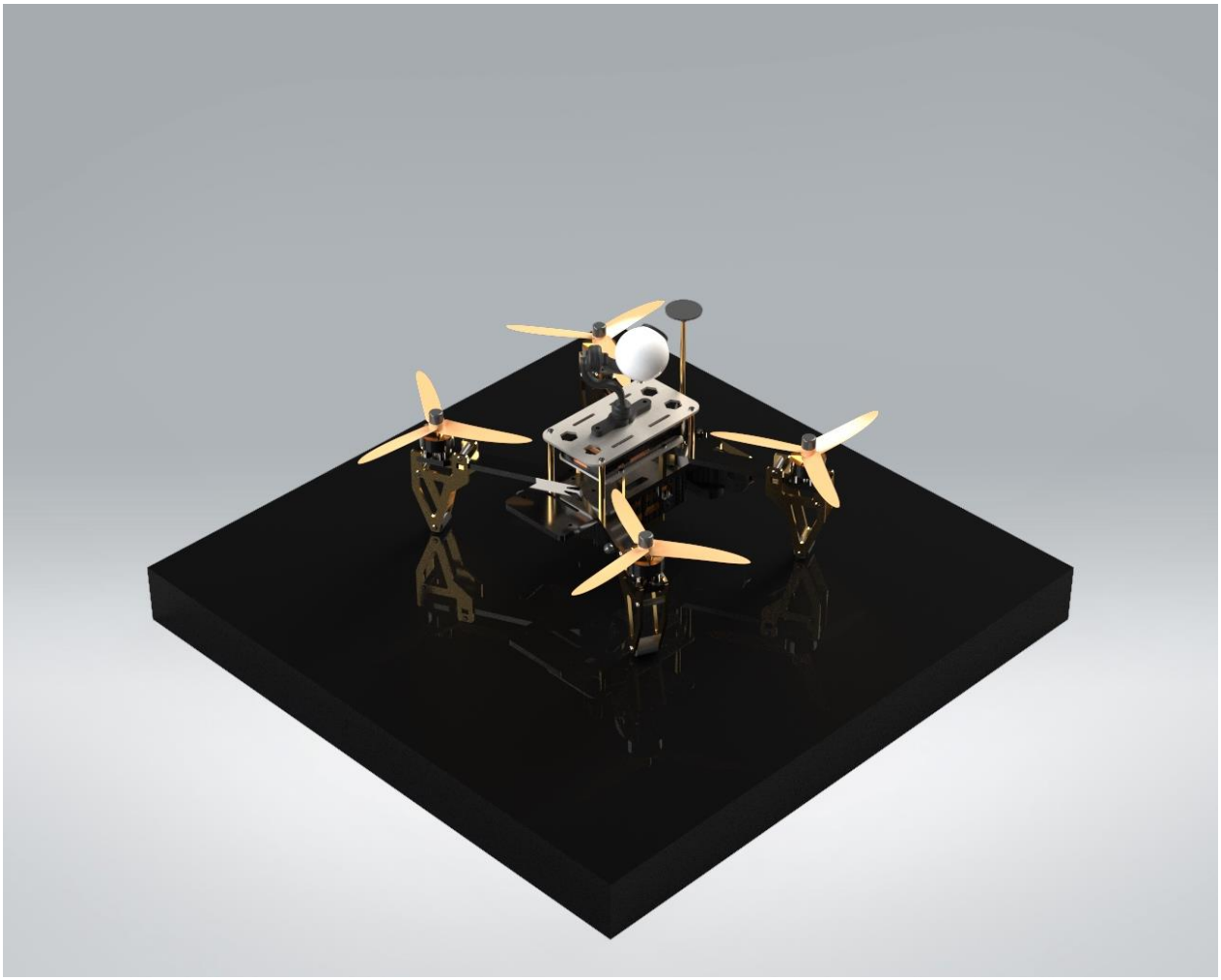


Figure 3: Full assembly Rendered.



Figure 4: Exploded View.

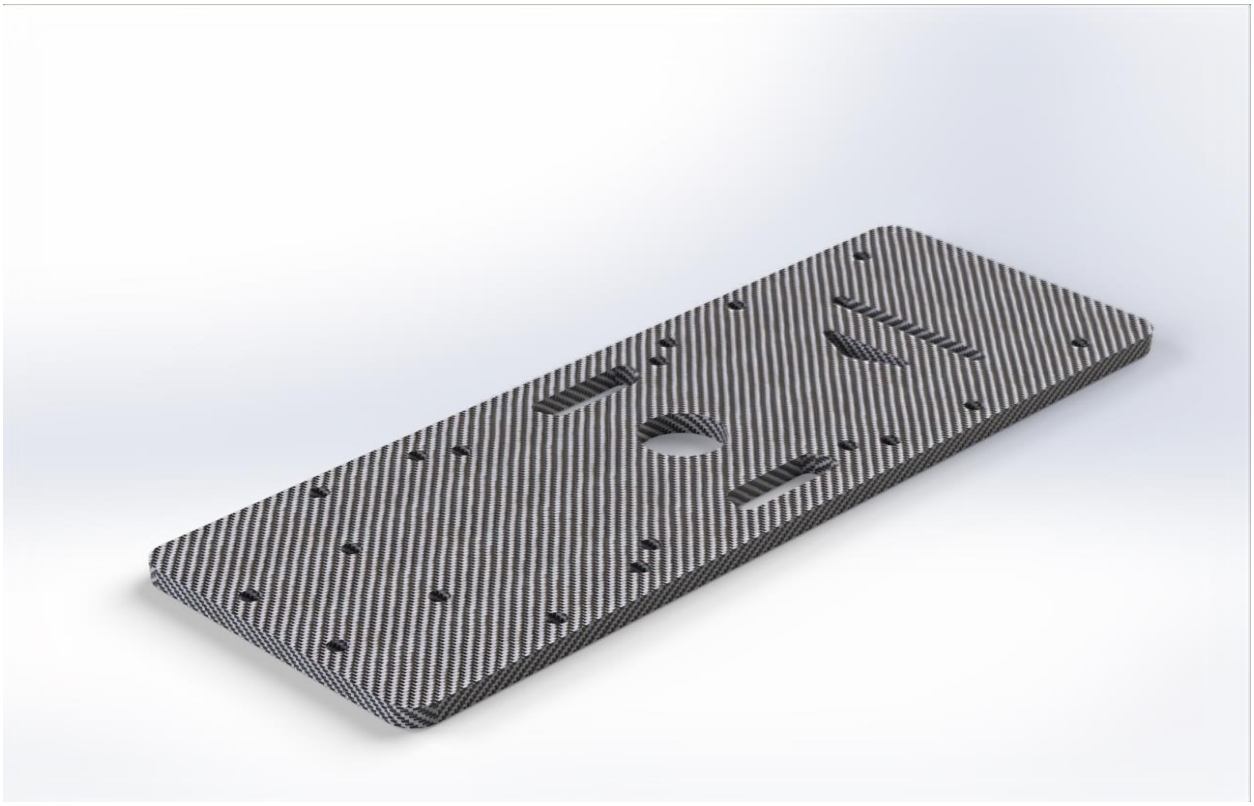


Figure 5:Main frame rendered.

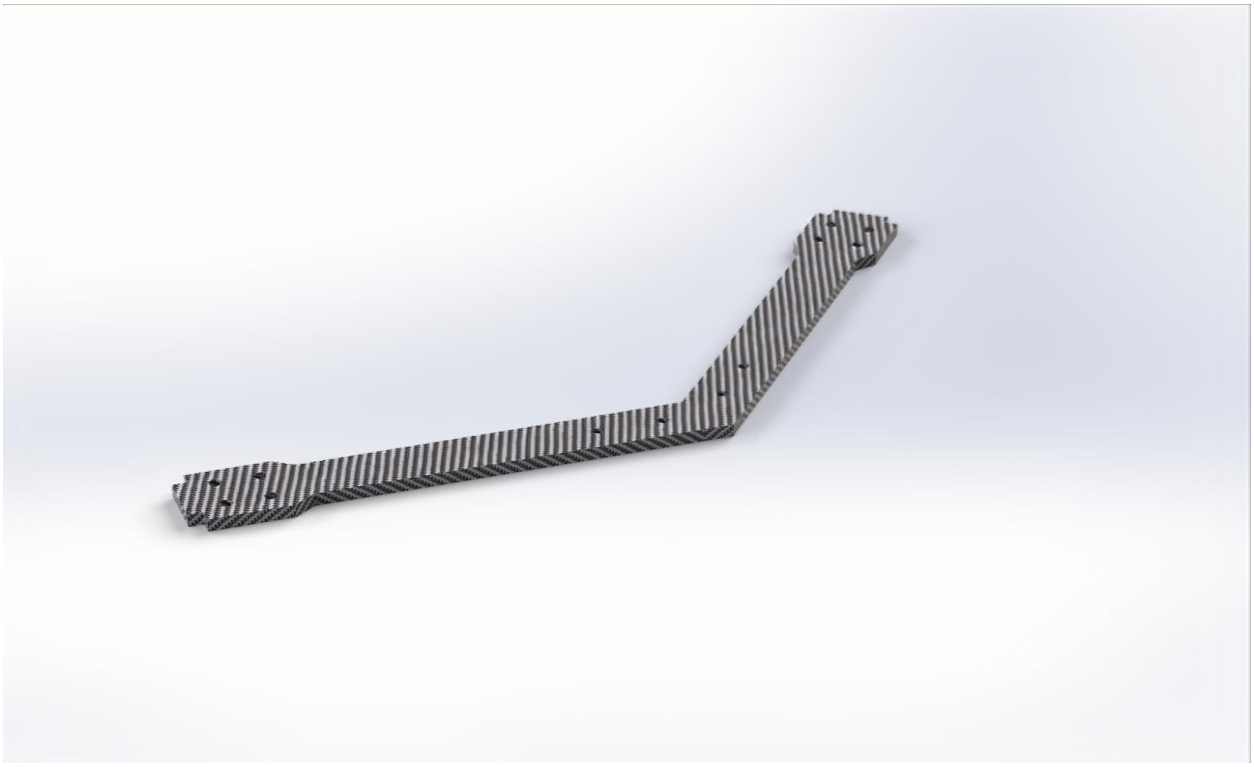


Figure 6:V bracket rendered

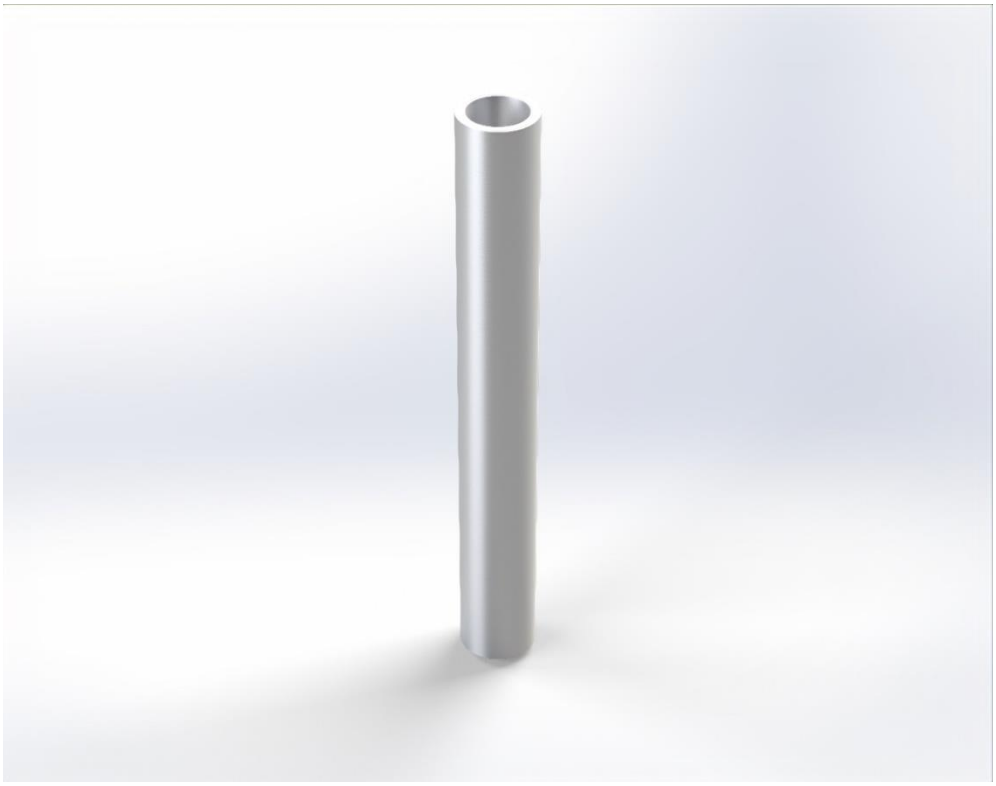


Figure 7:Battery Frame Spacer rendered.

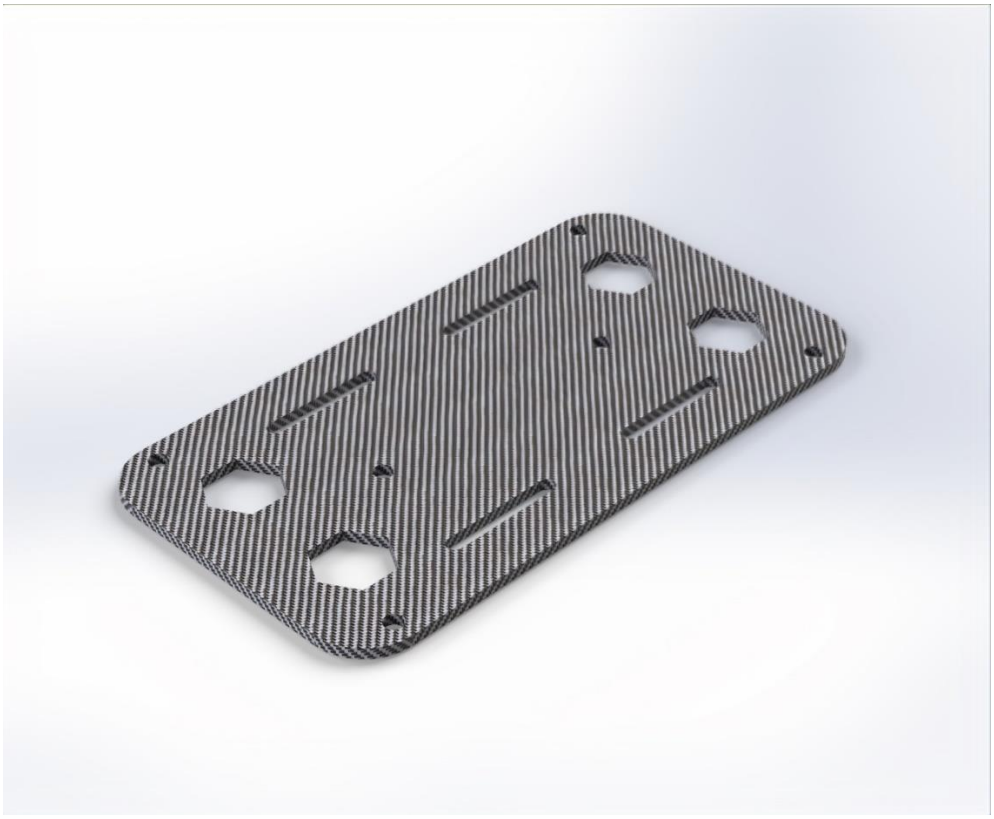


Figure 8:Battery frame rendered.

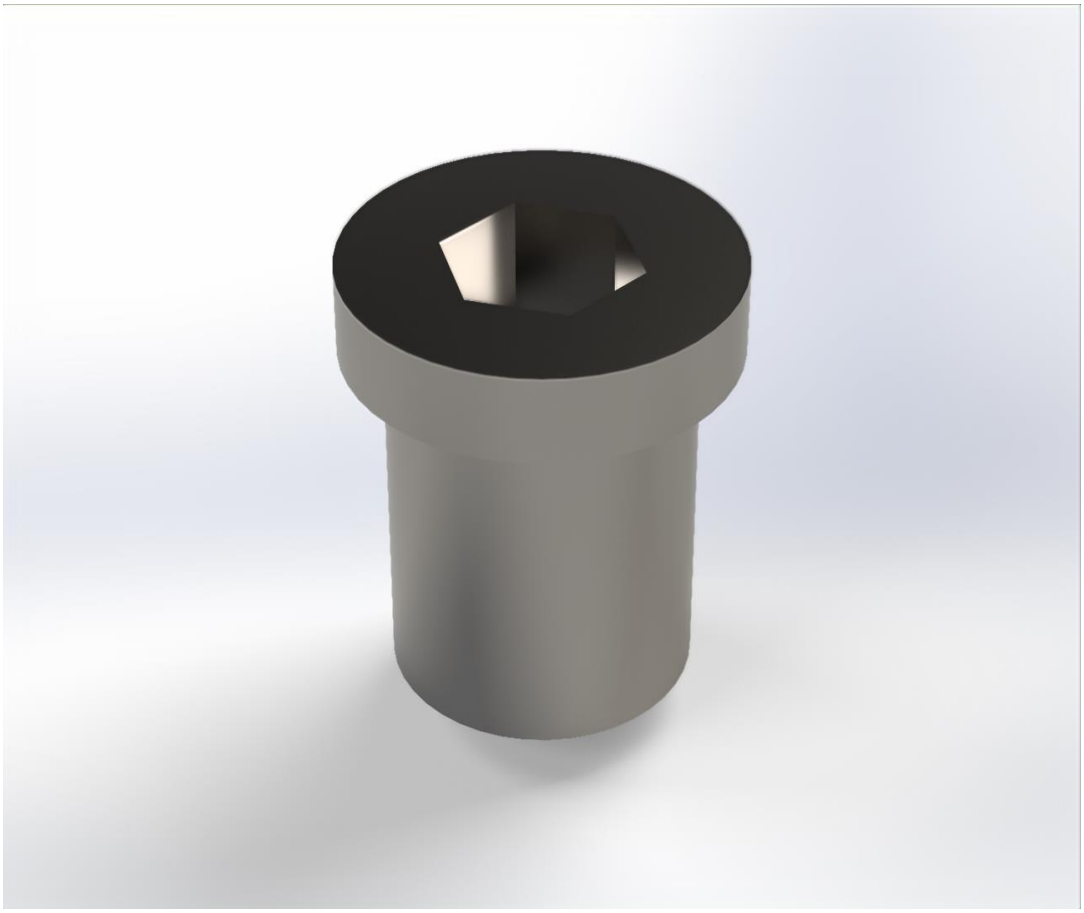


Figure 9:Screw battery Frame.

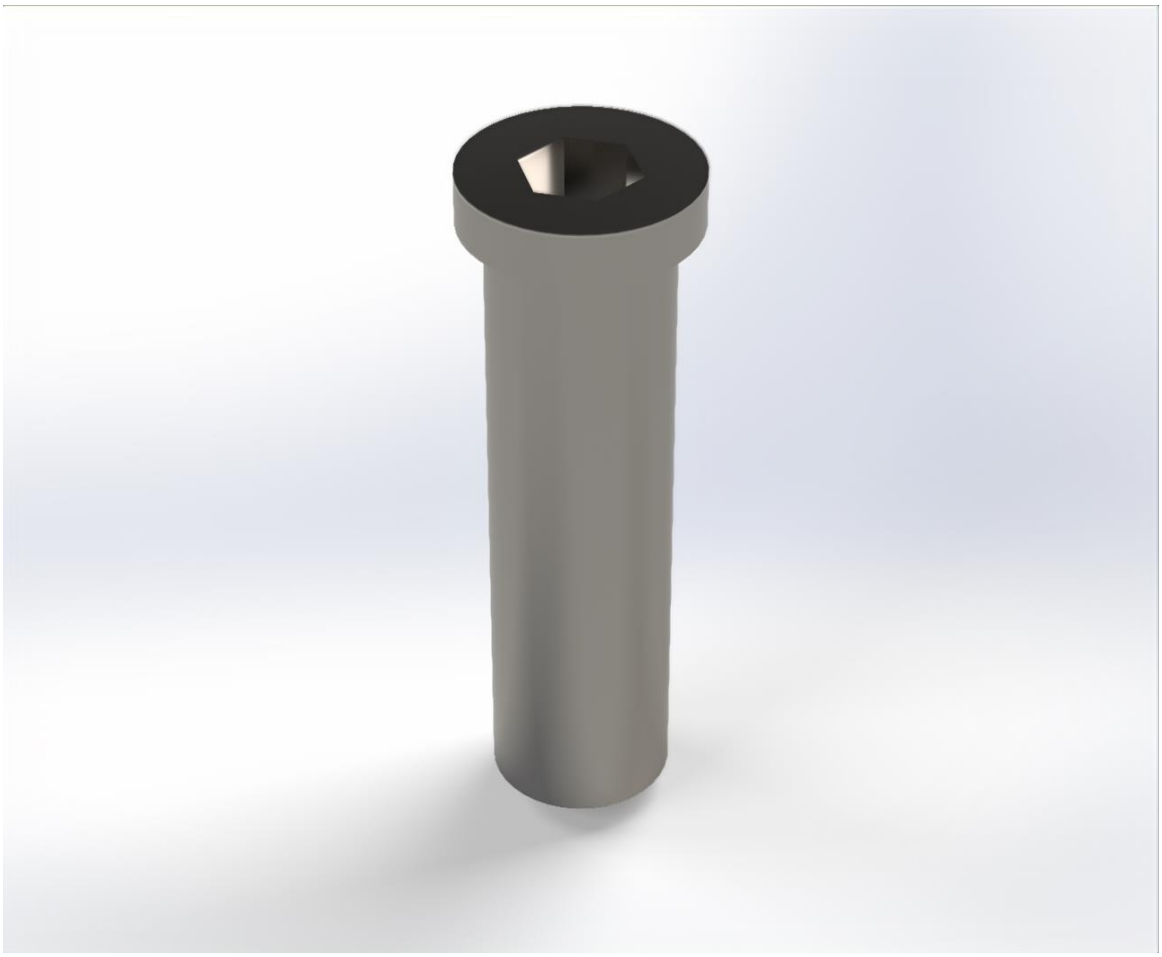


Figure 10:Screw V bracket Main frame.

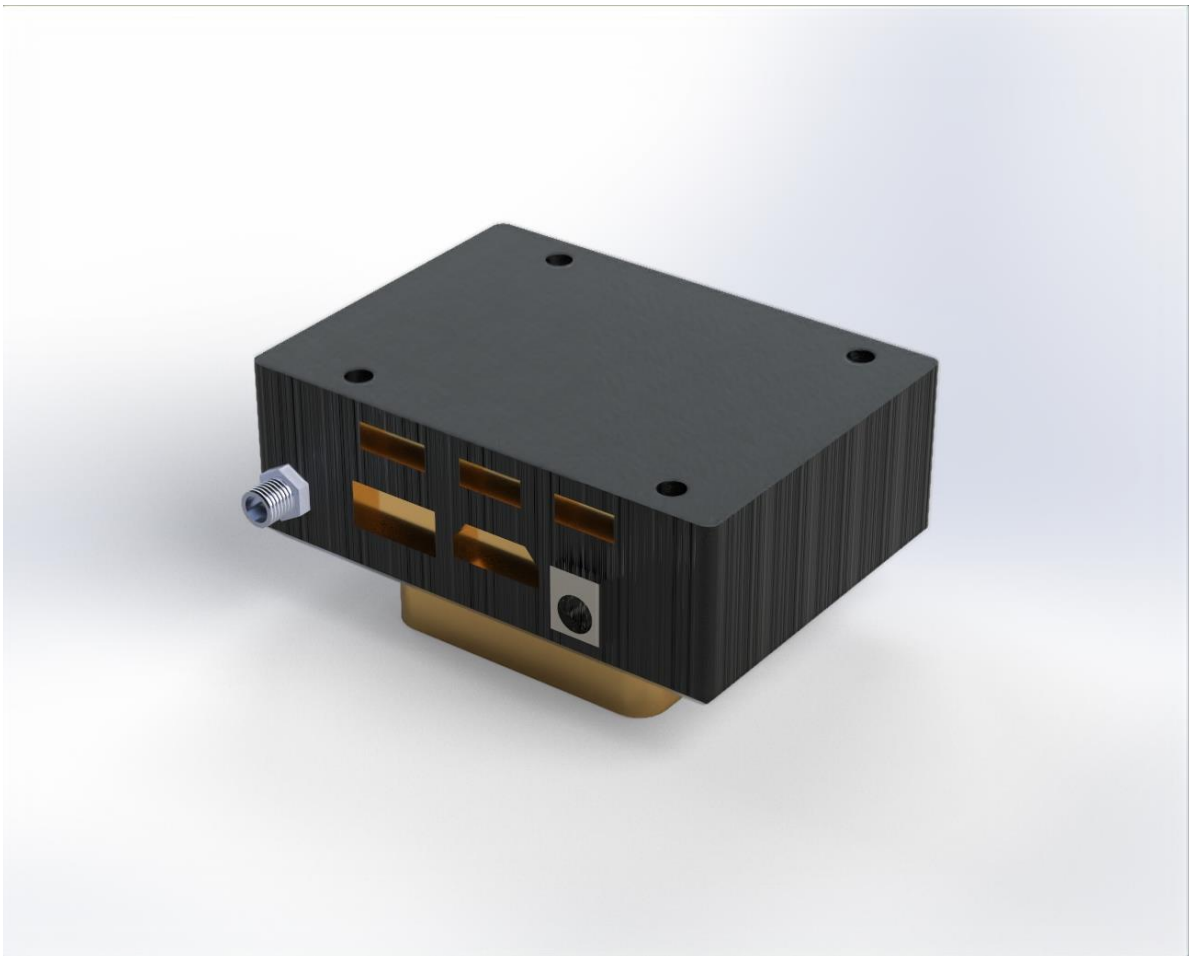


Figure 11:Companion computer V2 rendered.

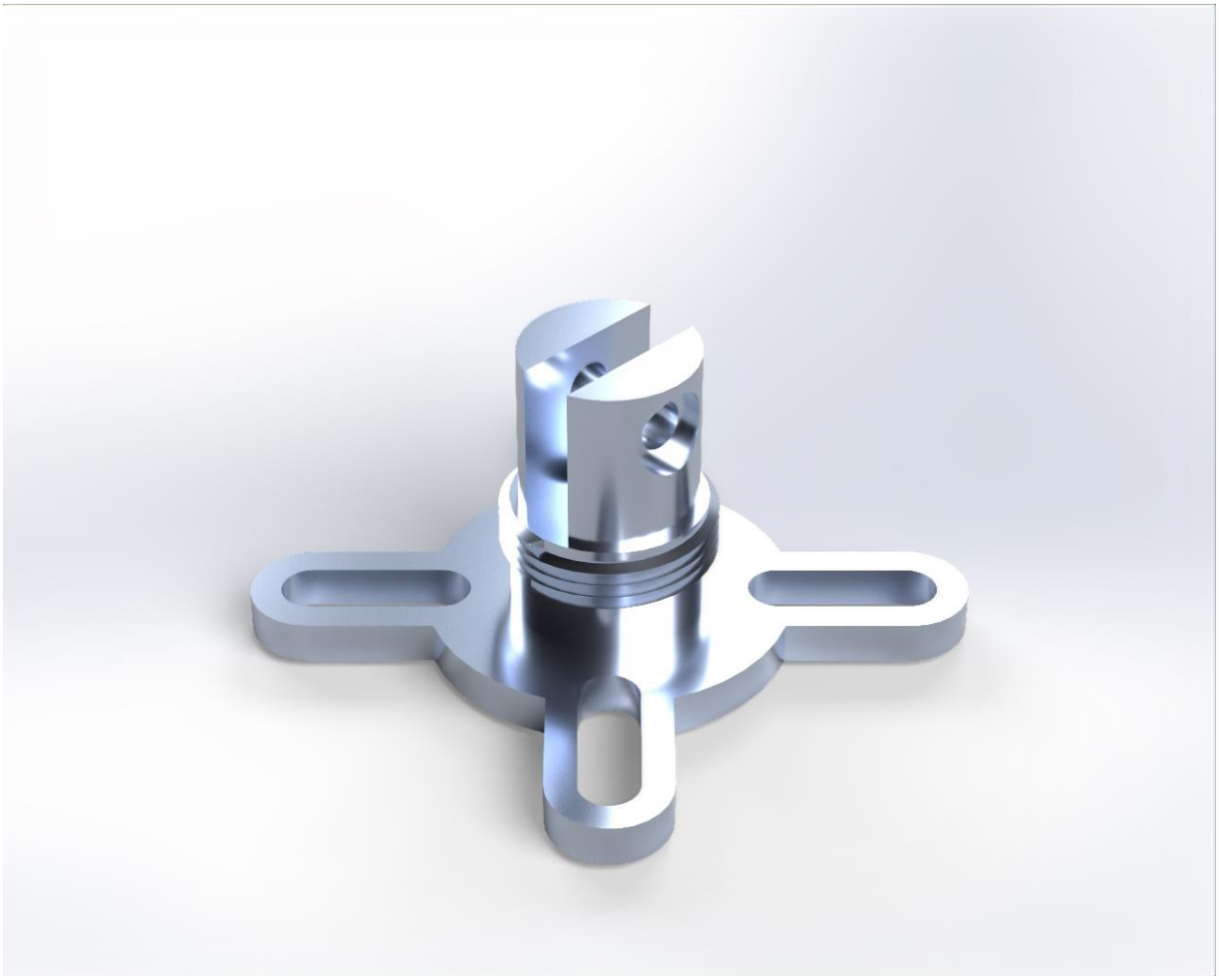


Figure 12:GPS base.

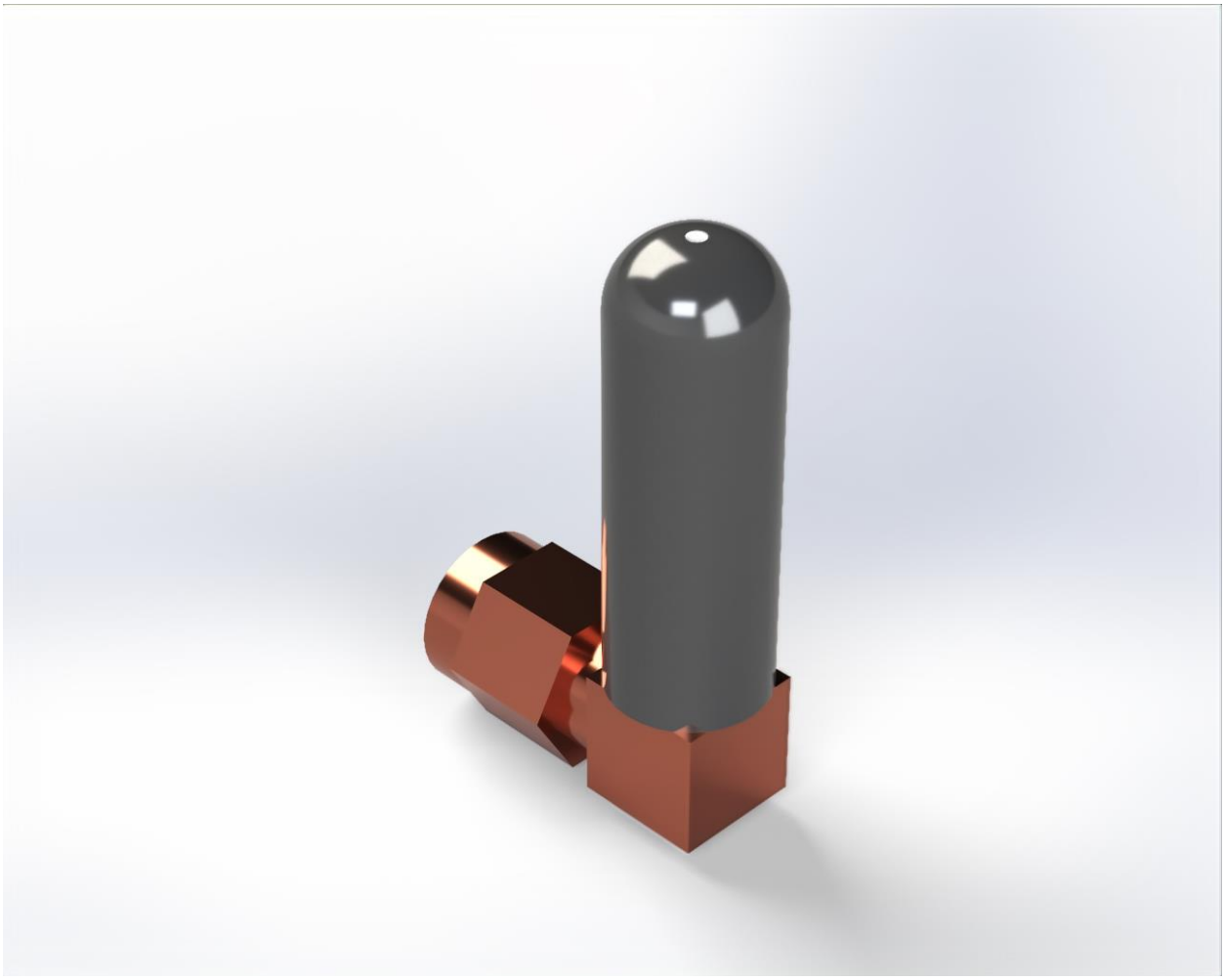


Figure 13:Companion computer Antenna.

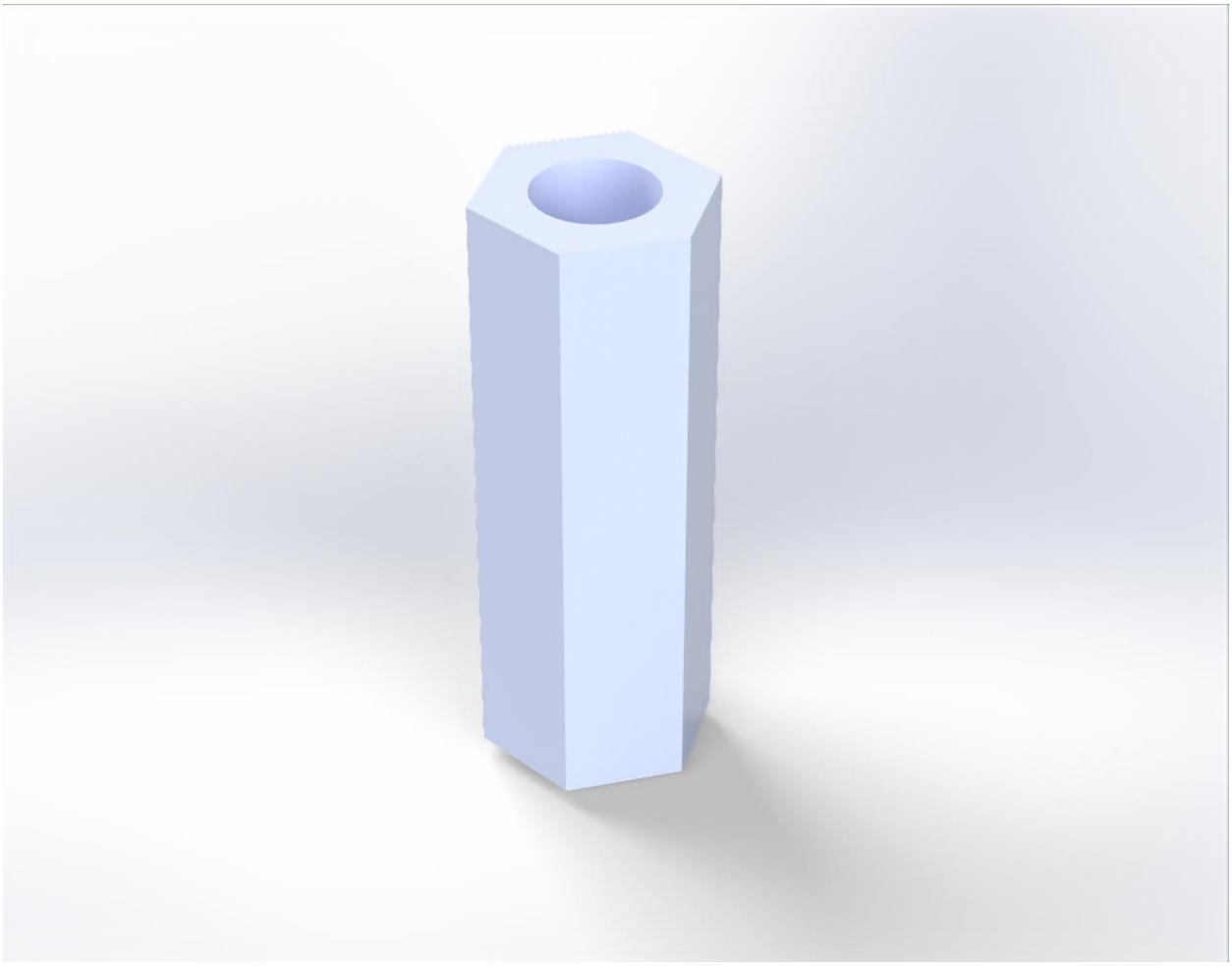


Figure 14:Flight controller frame Spacer.

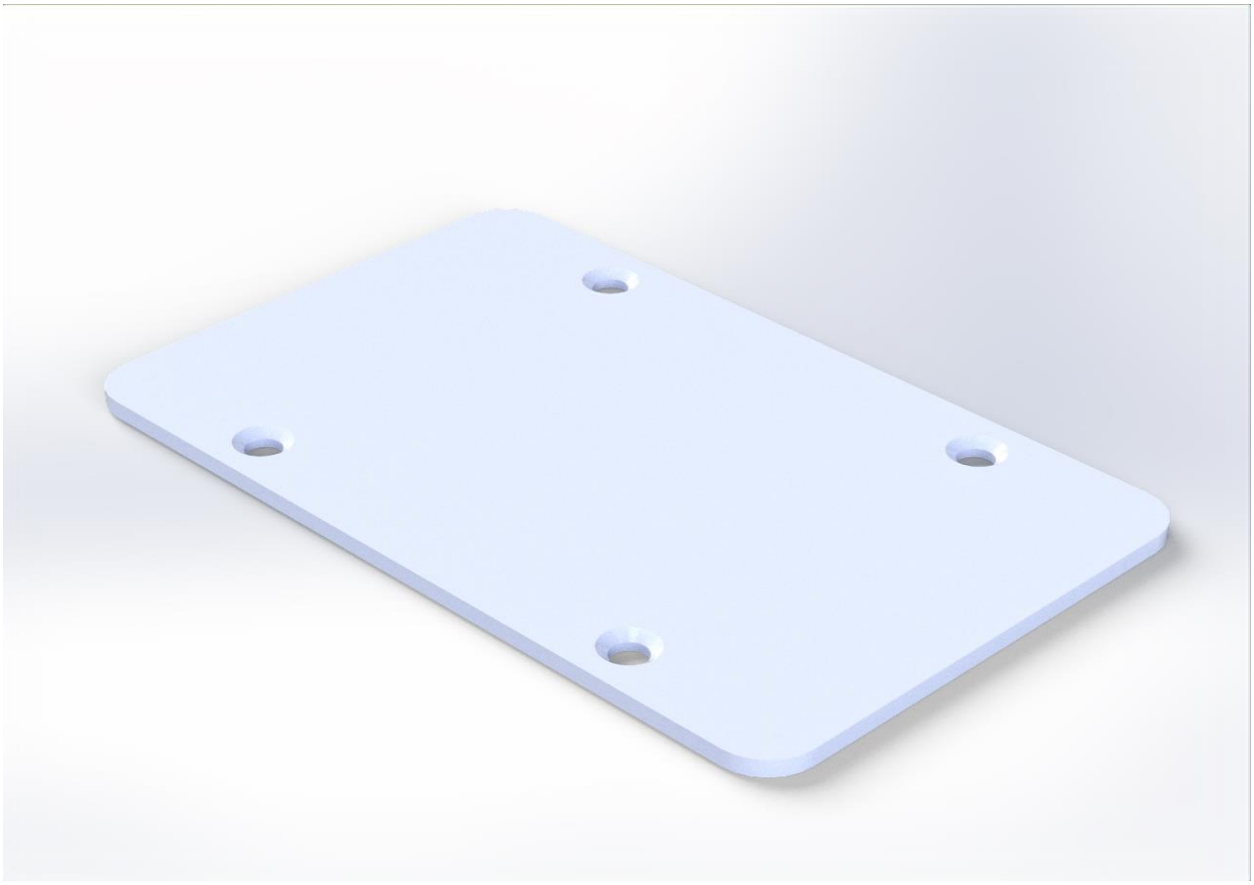


Figure 15:Flight Controller frame.

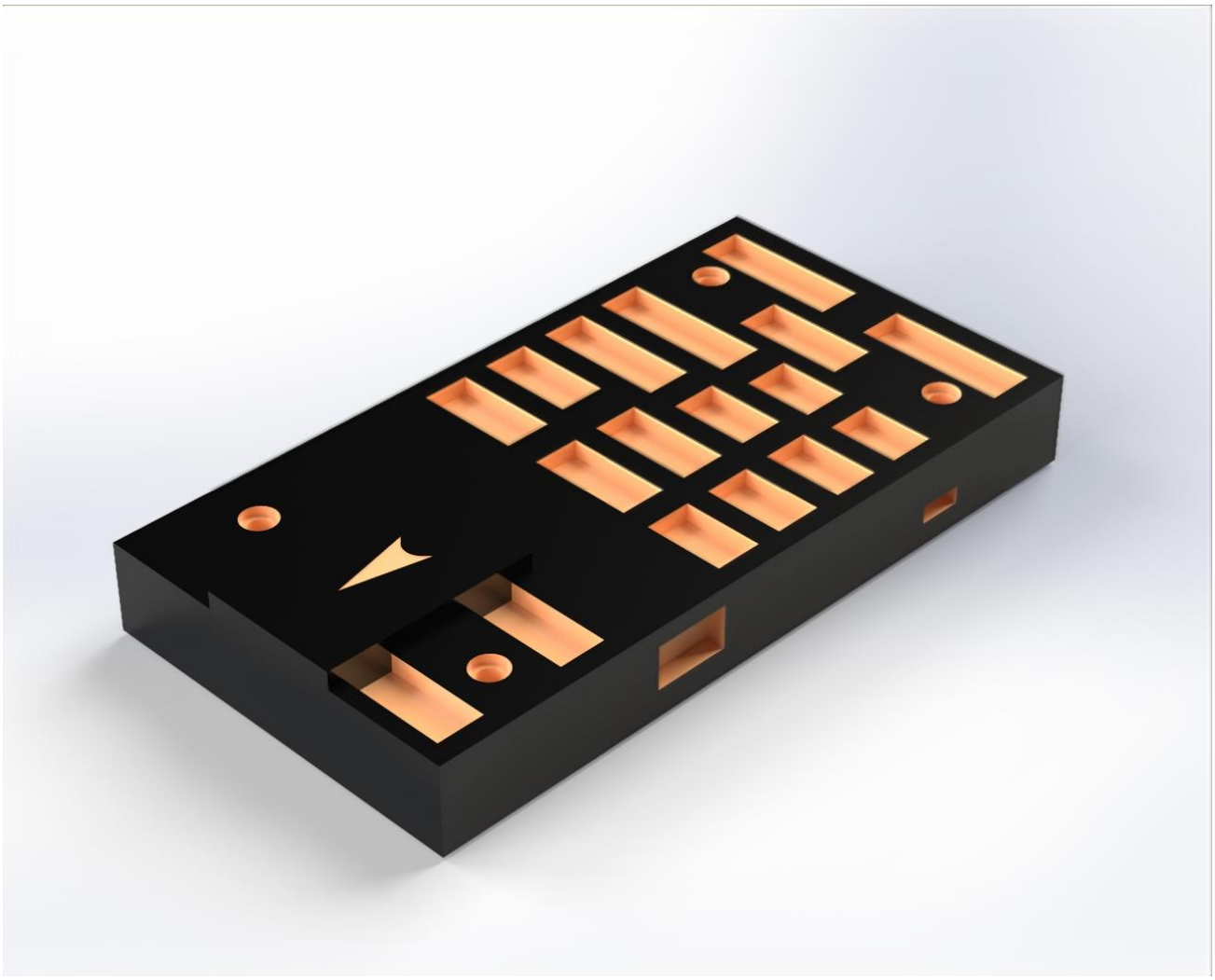


Figure 16: Pixhawk 4.



Figure 17:GPS screw cap.

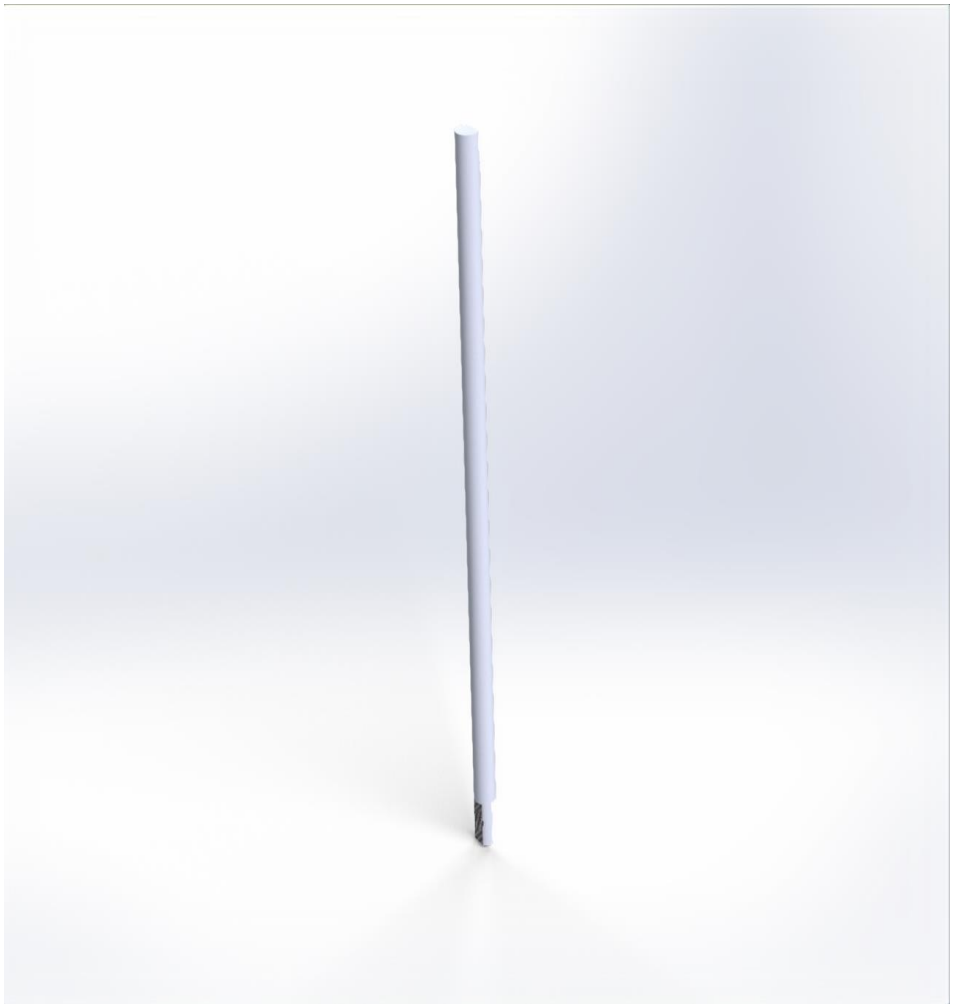


Figure 18:GPS long rod



Figure 19:GPS mount.



Figure 20:Pin for rod upper.

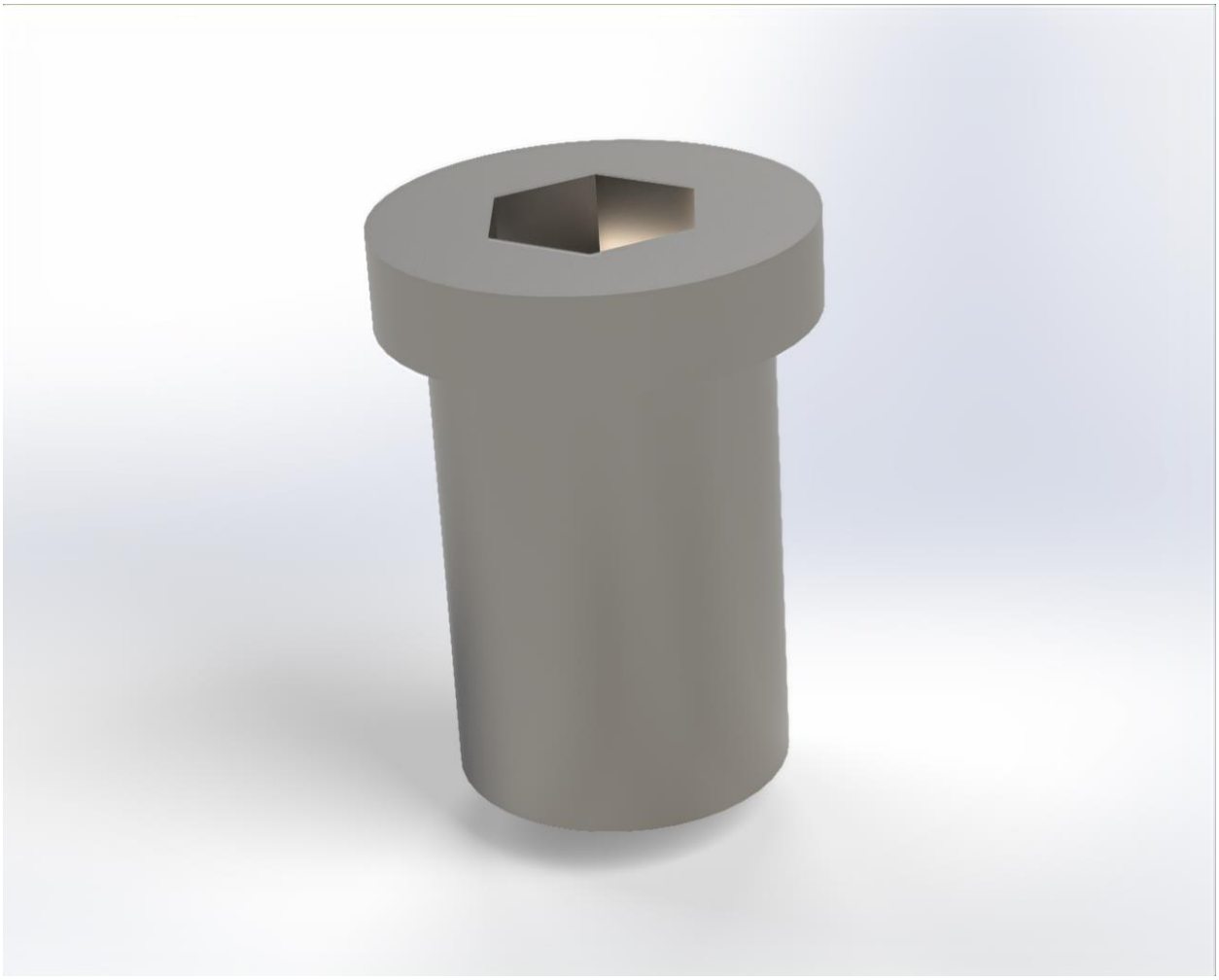


Figure 21:screw GPS base

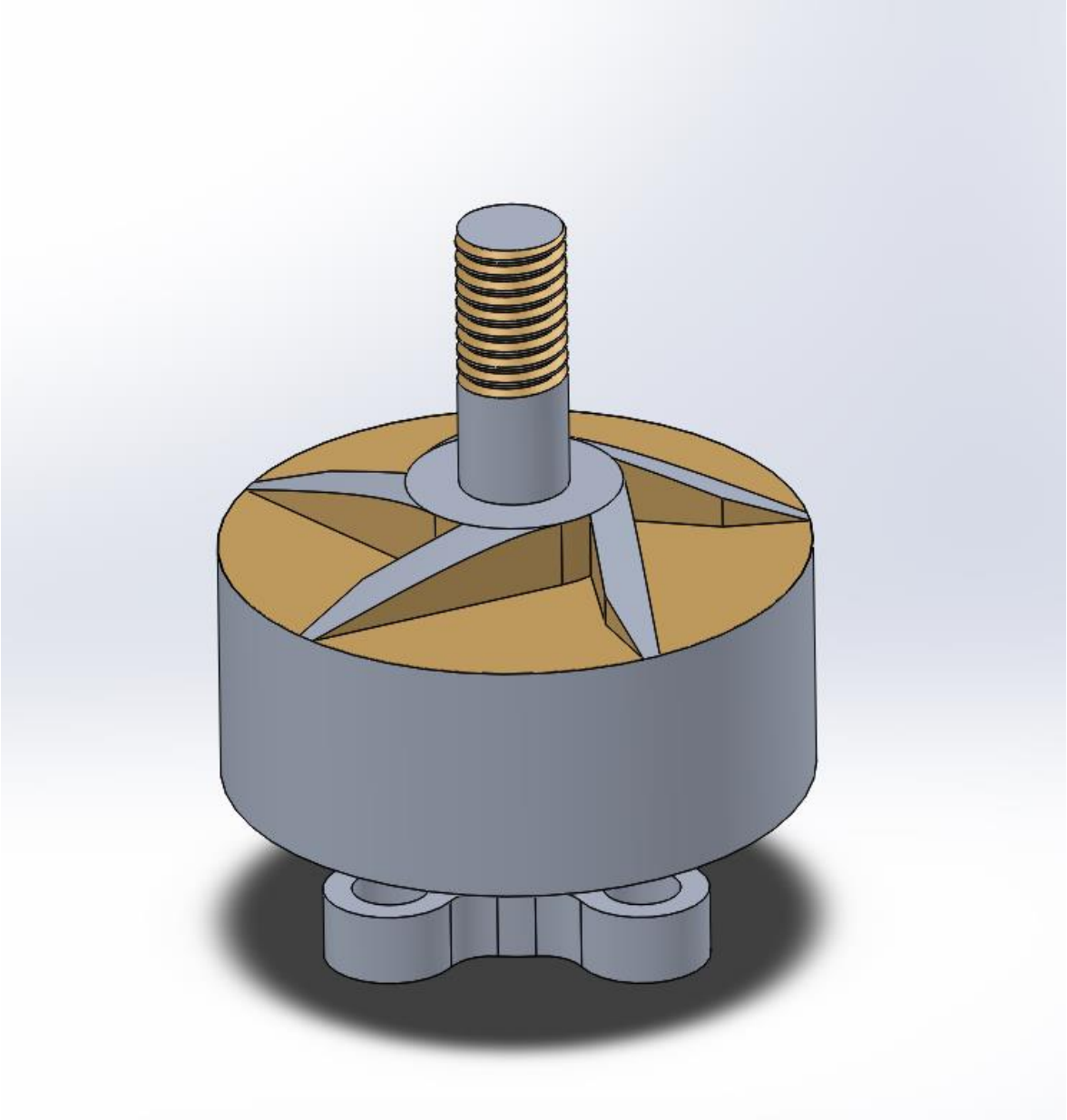


Figure 22:servo motor F60PRO IV V2.0 KV1750.

(this part was not rendered due to an error encountered while exporting to visualize).

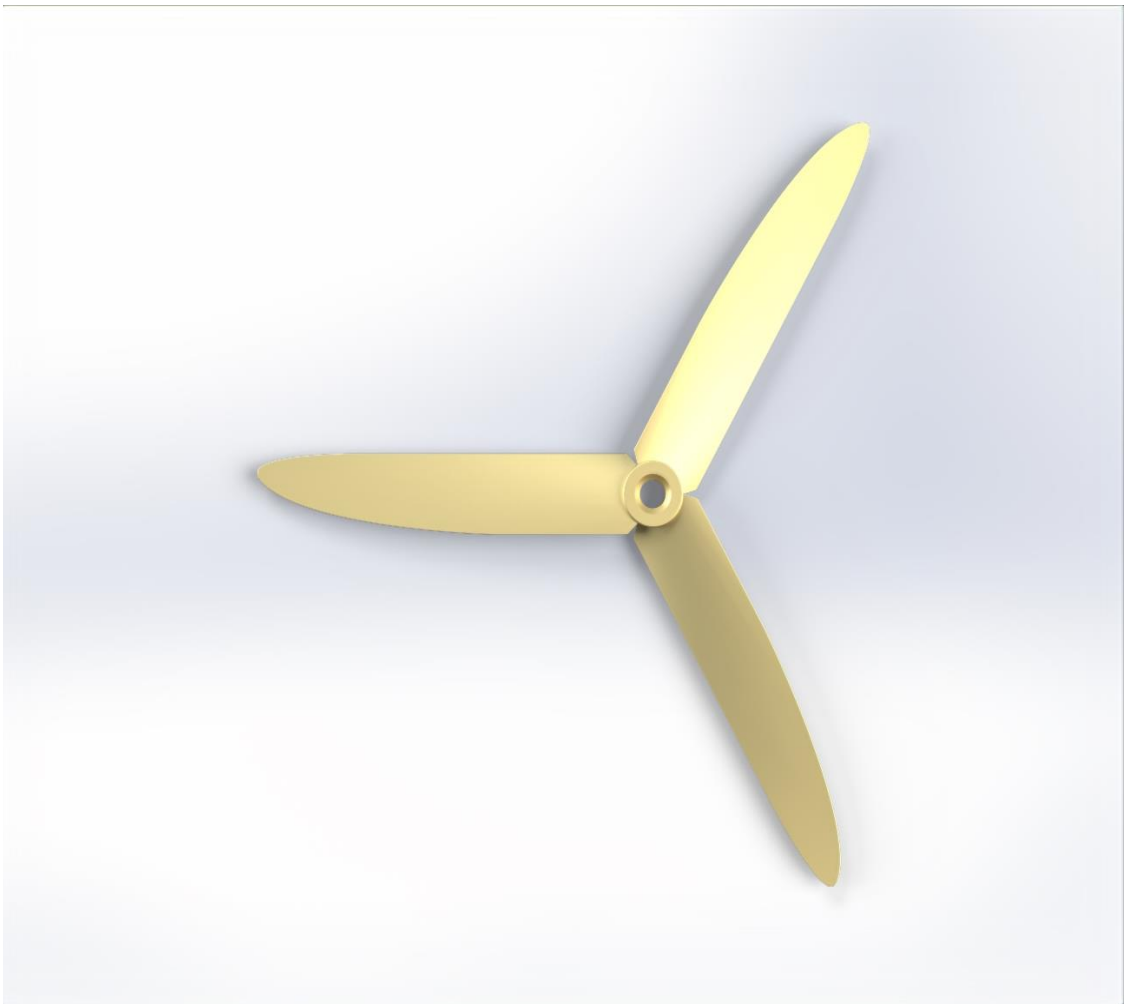


Figure 23:Propeller 6045R.

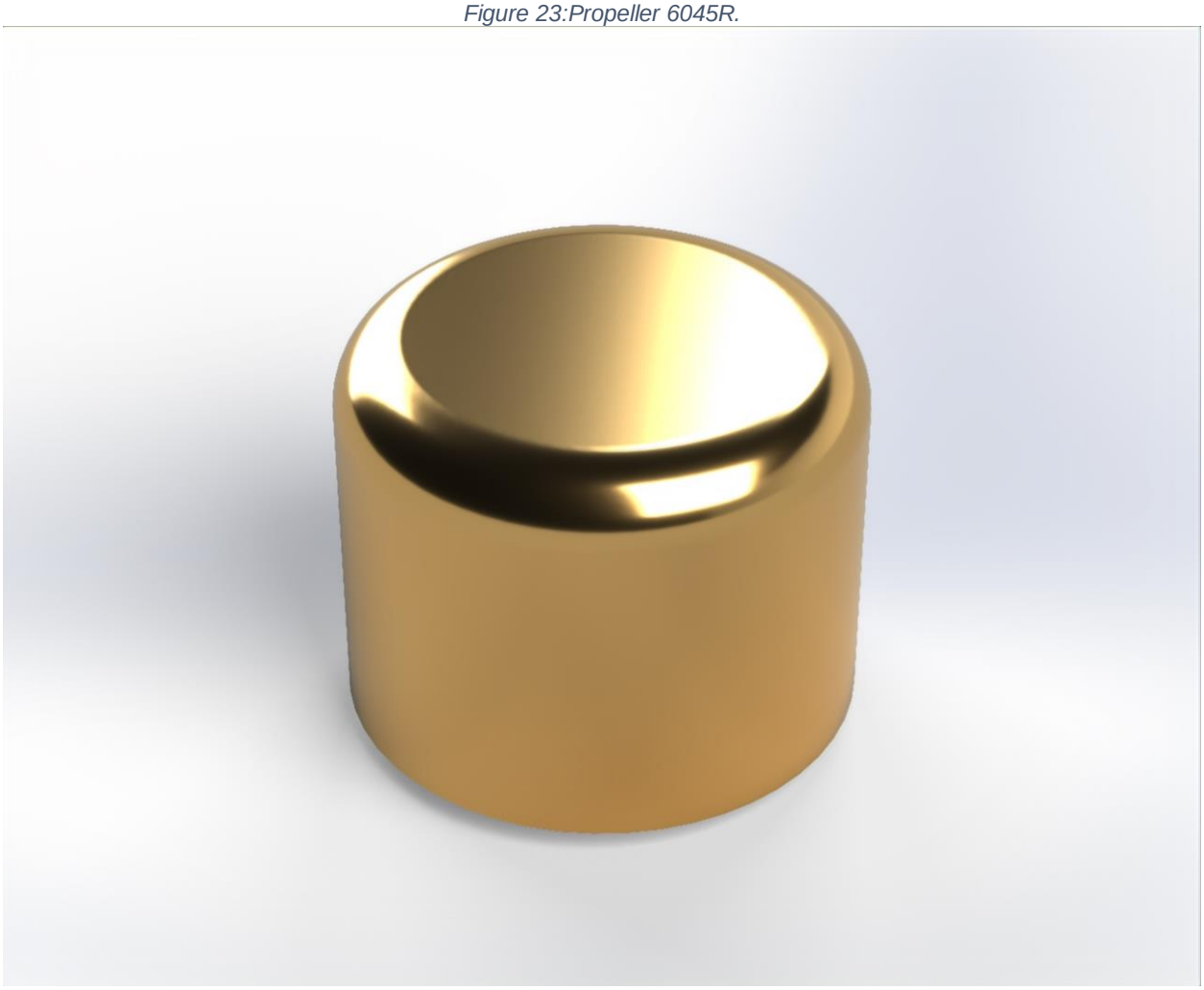


Figure 24:Servo motor cover.



Figure 25:Pin spacer thin.



Figure 26:Pin spacer thick

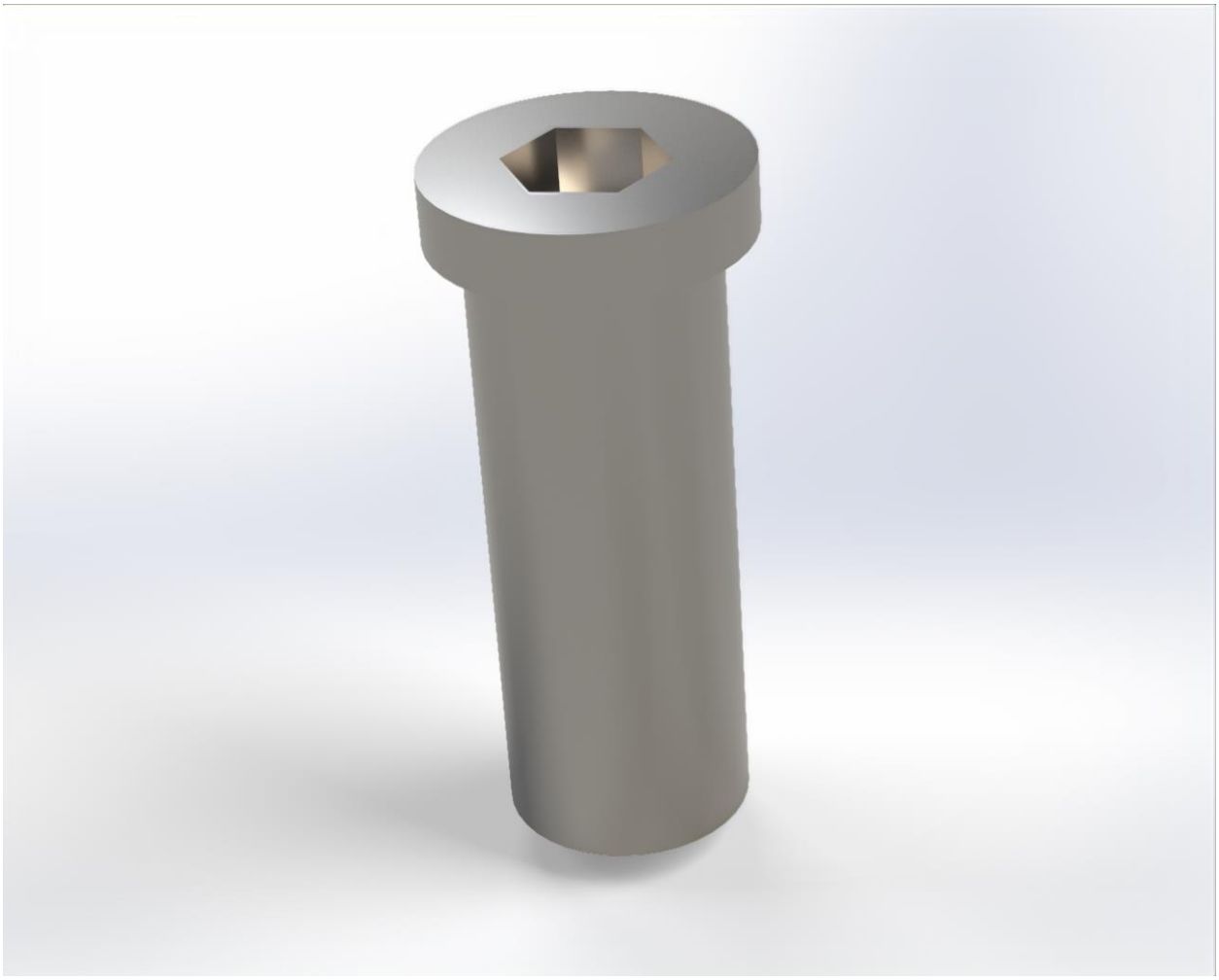


Figure 27:screw servo motor.

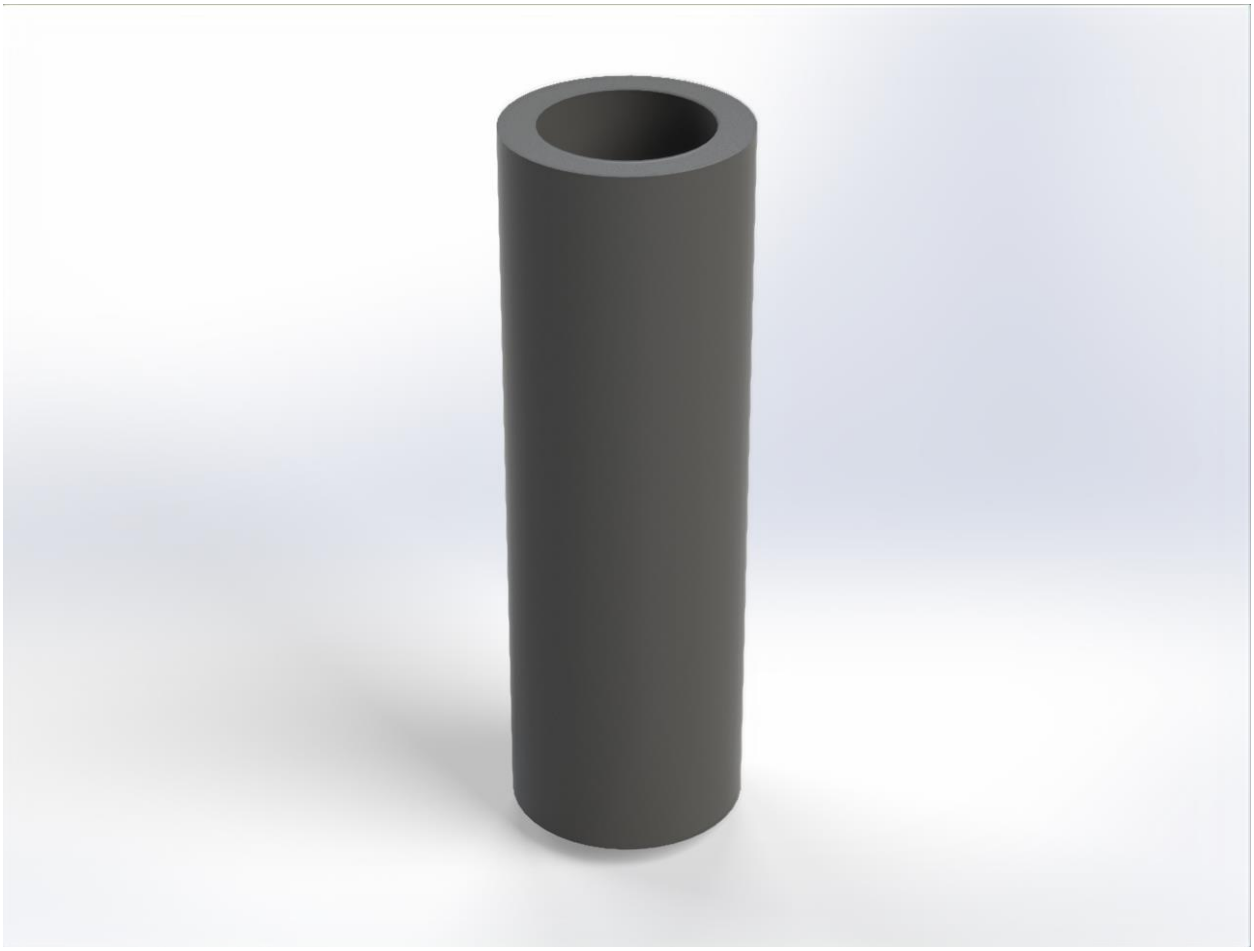


Figure 28:Landing leg spacer(thin)

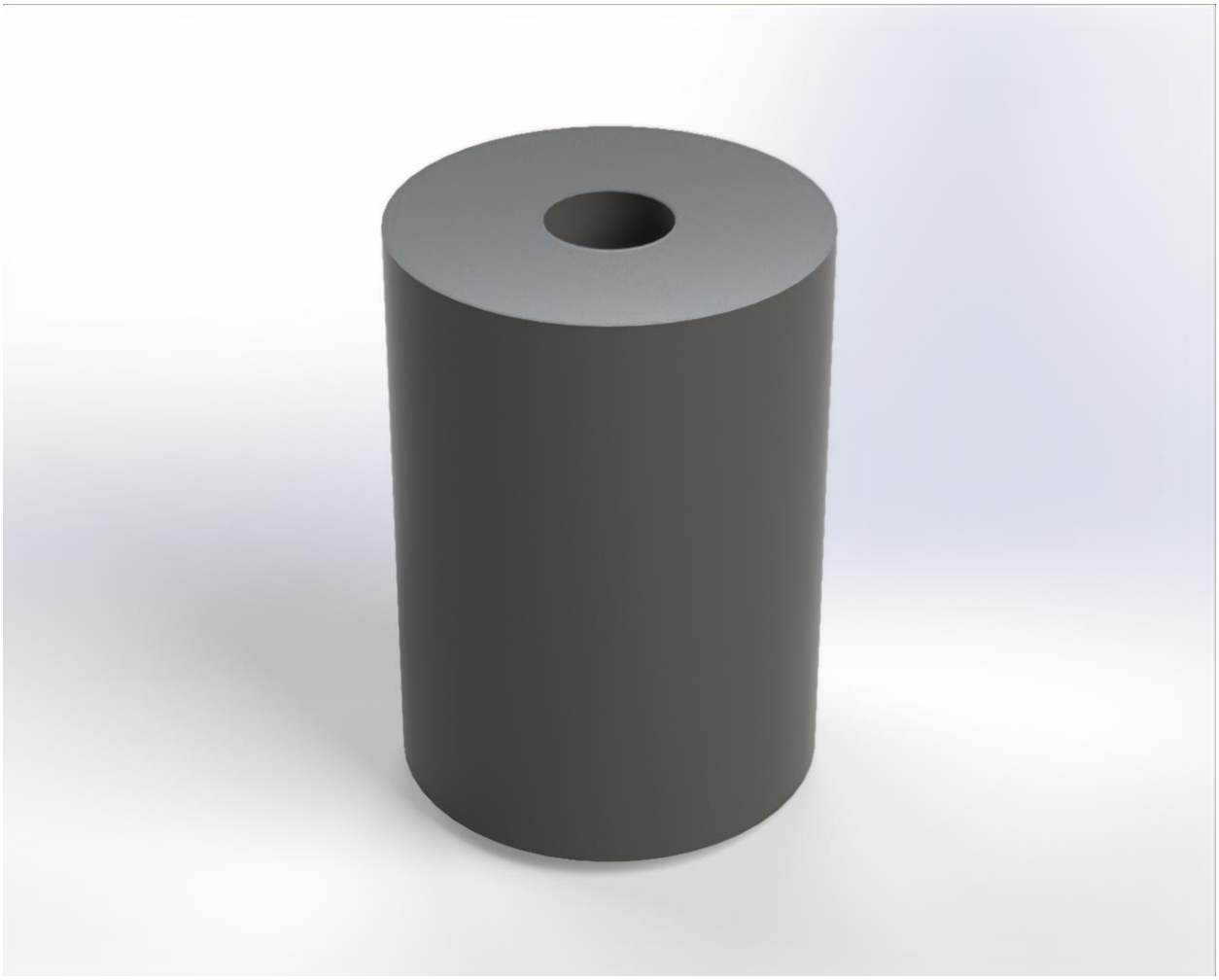


Figure 29:Landing leg spacer (thick)



Figure 30:Landing leg.

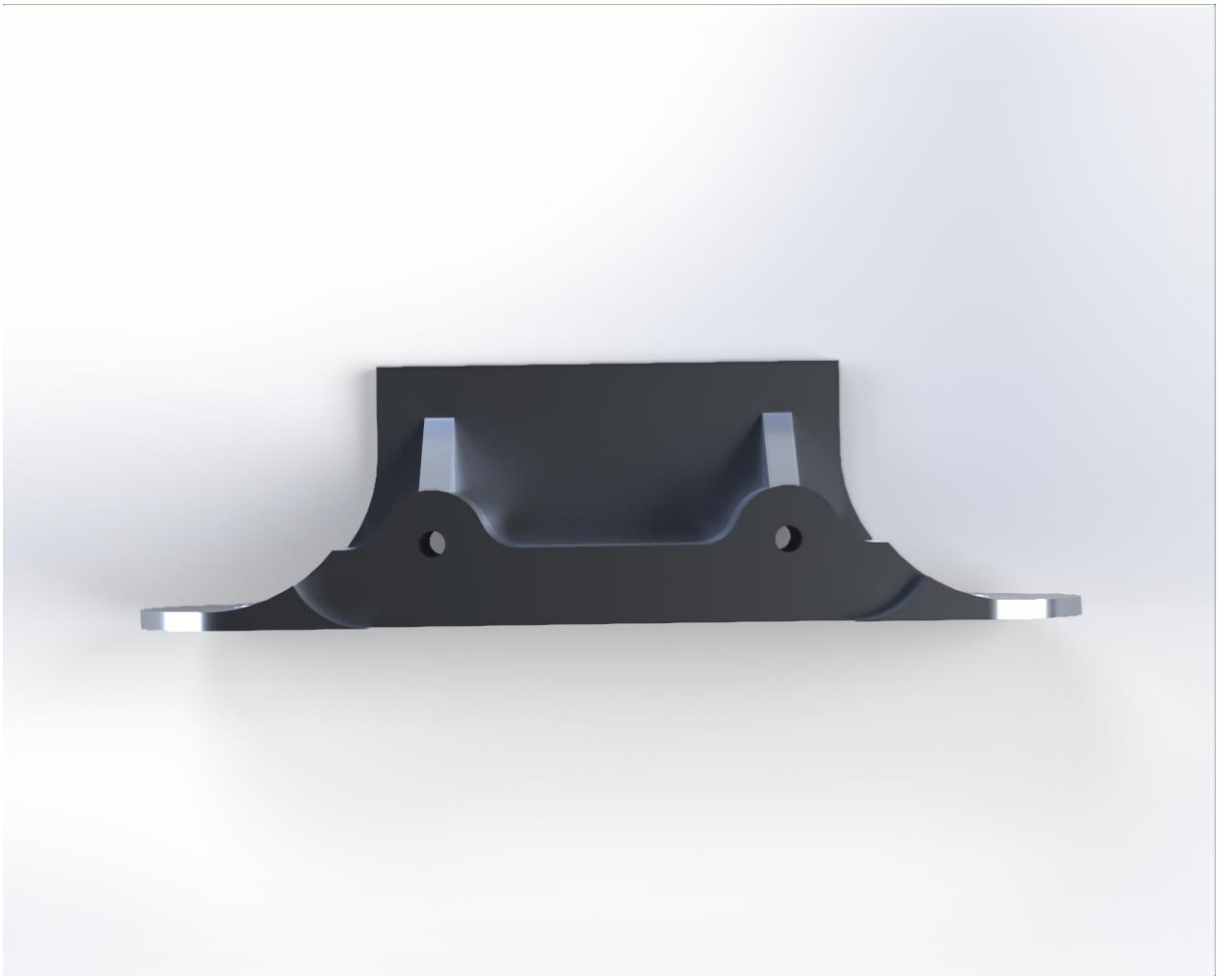


Figure 31:Structure core frame.



Figure 32:LIDAR mount.



Figure 33:LIDAR

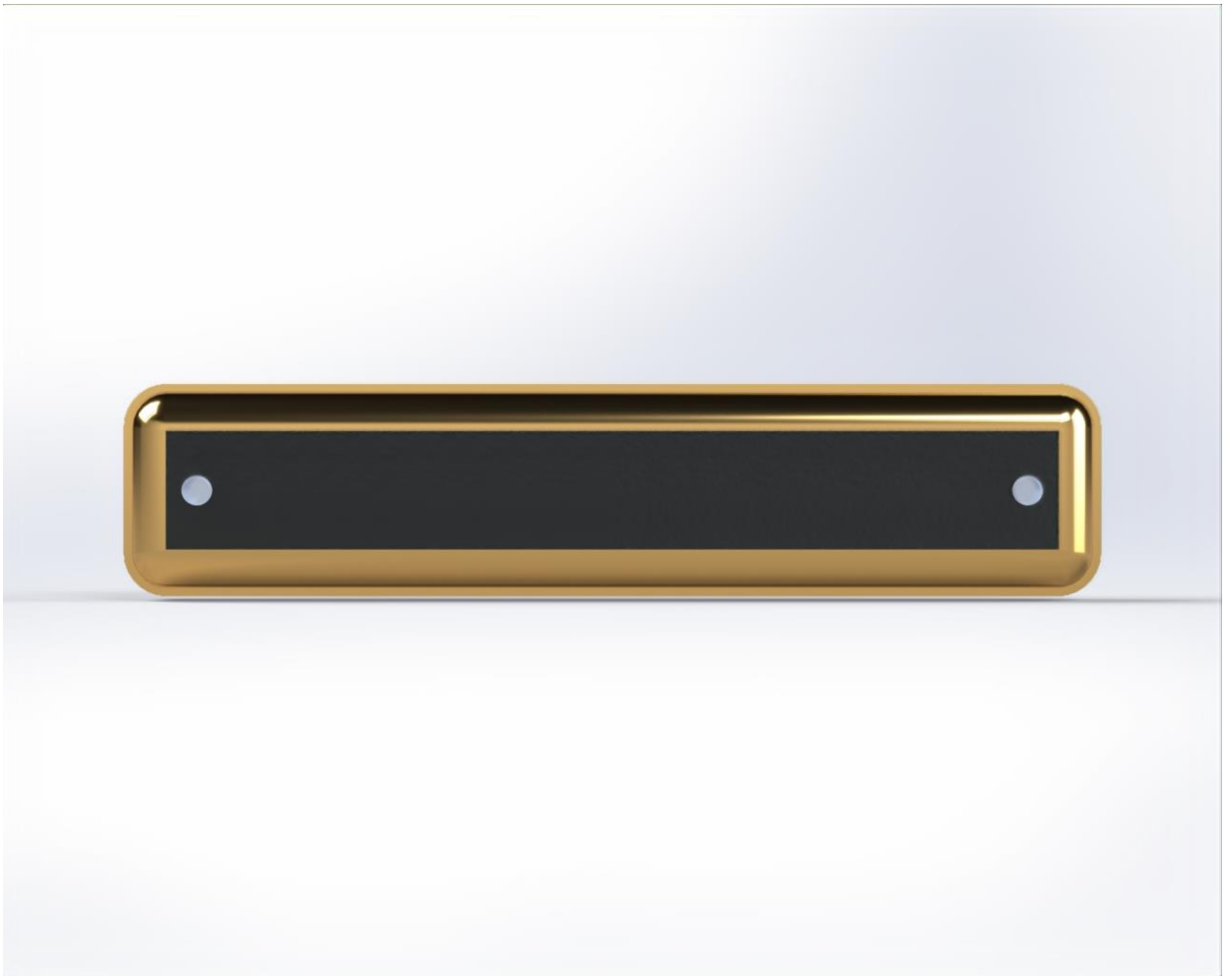


Figure 34:Frame T265

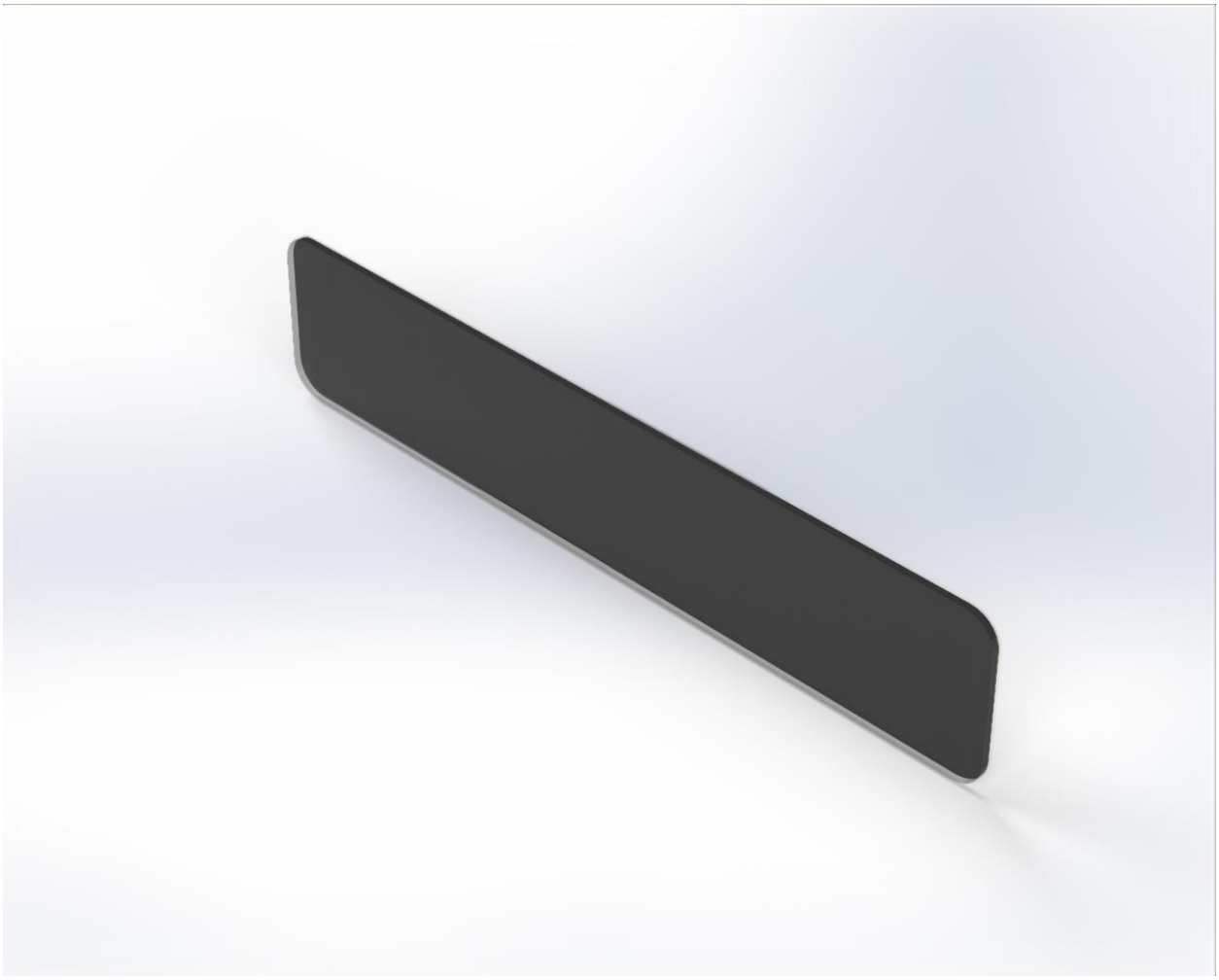


Figure 35: Screen T265.



Figure 36: Camera Assembly.

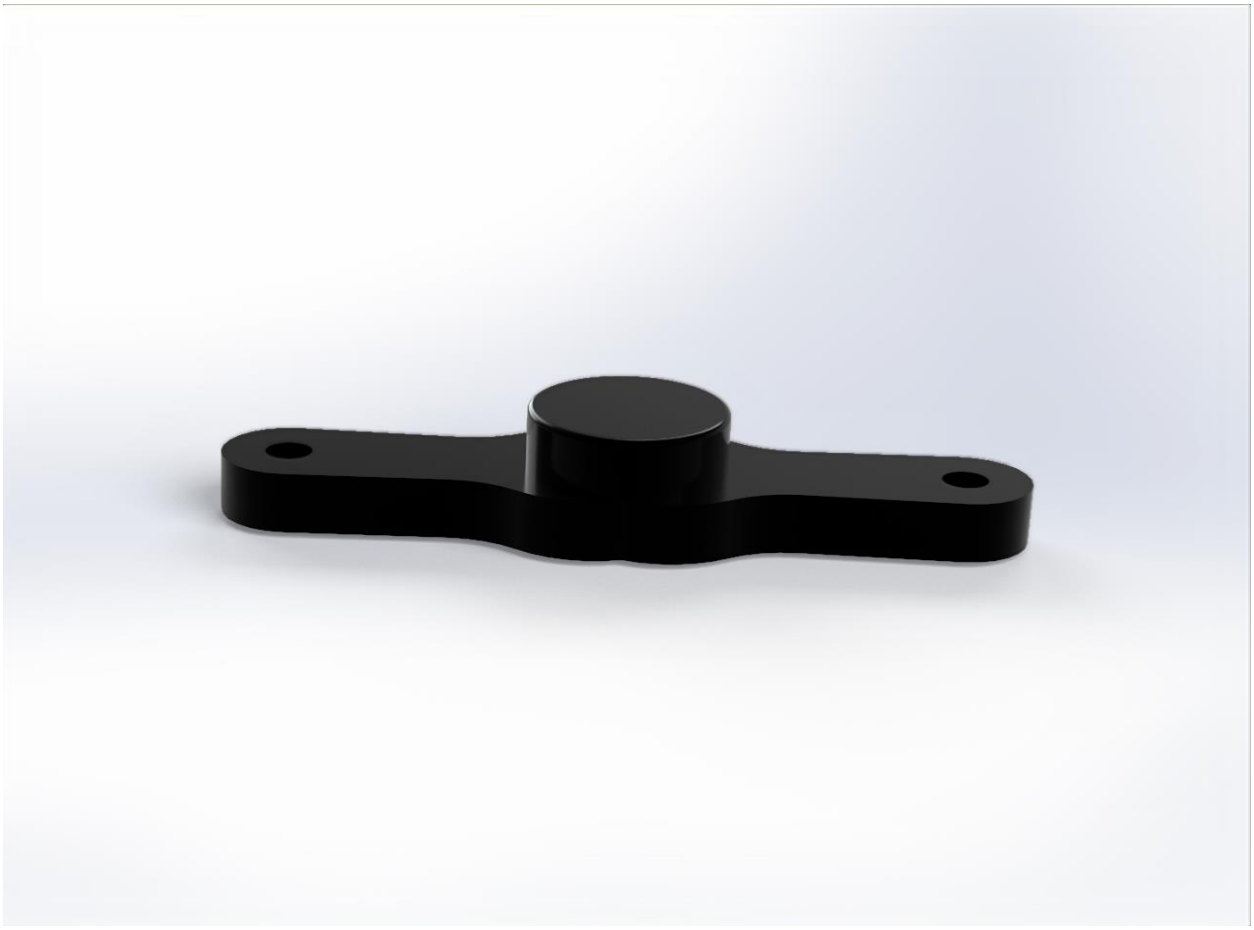


Figure 37:Camera arm base.



Figure 38:Camera arm.

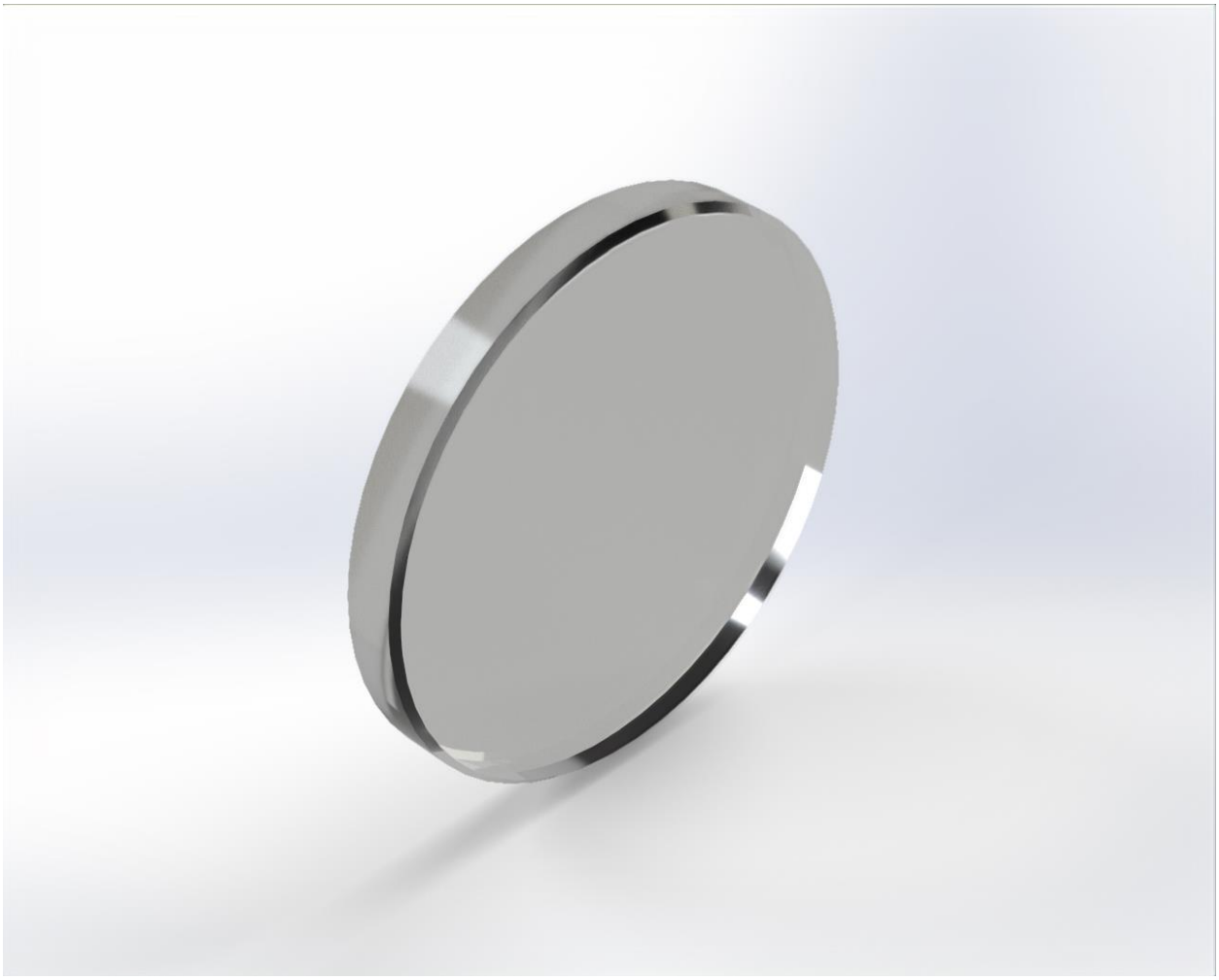


Figure 39: Camera lens.

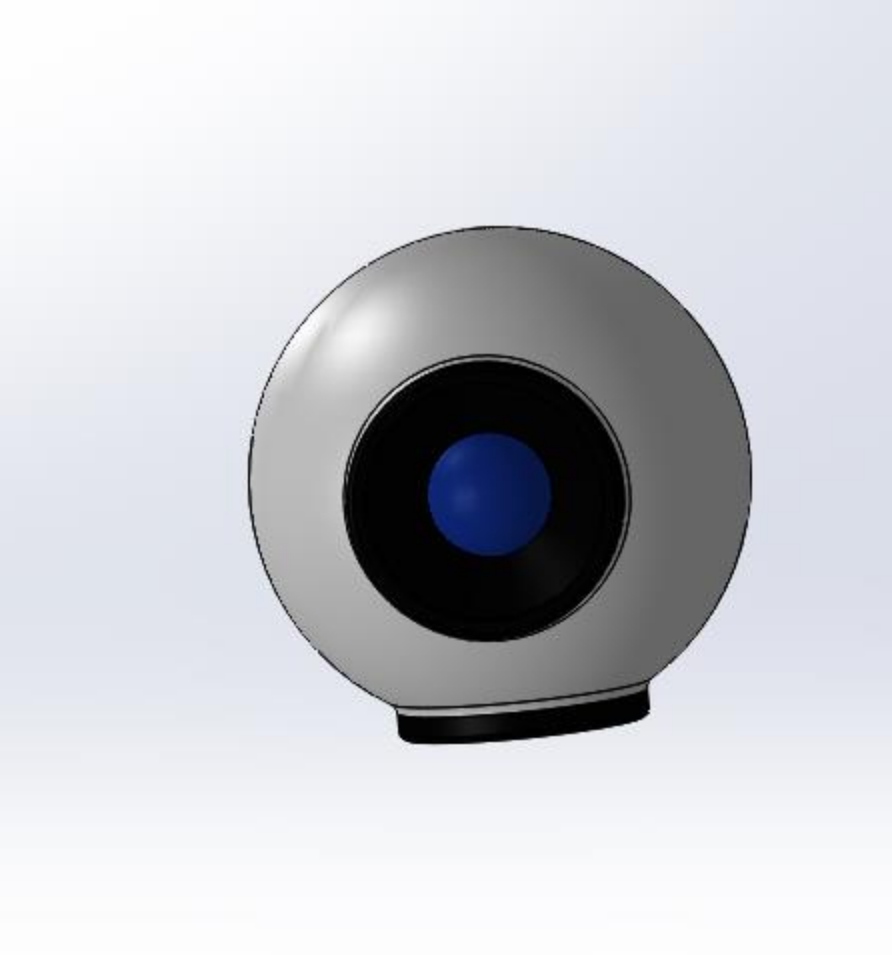


Figure 40:Camera sphere.

(this part was not rendered due to the same error encountered in the servo motor part)



Figure 41:Camera base arm.



Figure 42:GPS

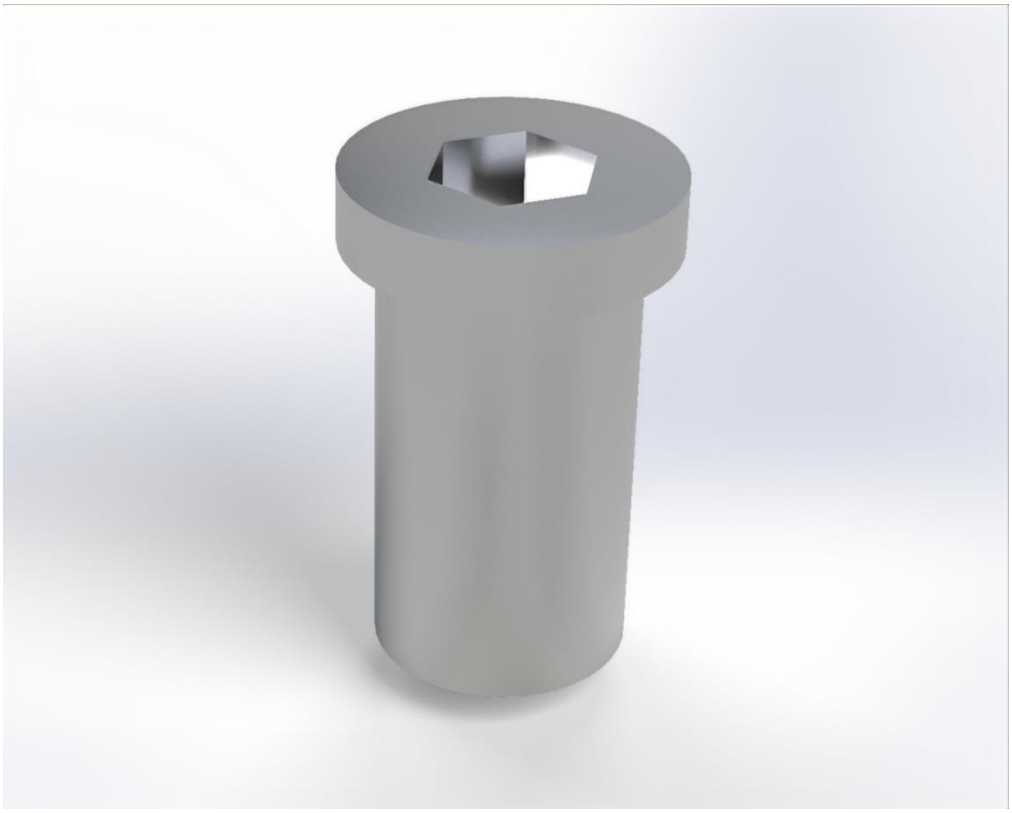


Figure 43: Screw Core Frame.



Figure 44: Pin for GPS Base.

Motion analysis:

We start off by calculating the force needed to lift the drone off the ground:

-The total weight of the drone was approximately 1255g, to make the drone lift off the ground the thrust force generated by the 4 motors must exceed the weight of the drone → F_{thrust}>12.31N

$F_{/motor}$=3.0755N to start lifting off the ground, we used 5N/Motor in our simulation to accelerate the drone upwards. Now to calculate the RPM of each motor to obtain a thrust force of 5N/Motor:

Our drone is a quadcopter having 3 blades/propeller, a typical thrust coefficient for the 3-bladed 6045R is C_t=0.14 → $n = \sqrt{\frac{F}{Ctx\boldsymbol{\rho}(\boldsymbol{air})\boldsymbol{xD}(\boldsymbol{propeller}\text{^})\mathbf{4}}}$ where D is the diameter of the propeller which is 0.14m in our case→n=275 RPS→n=16530 RPM

After applying these conditions and simulating:

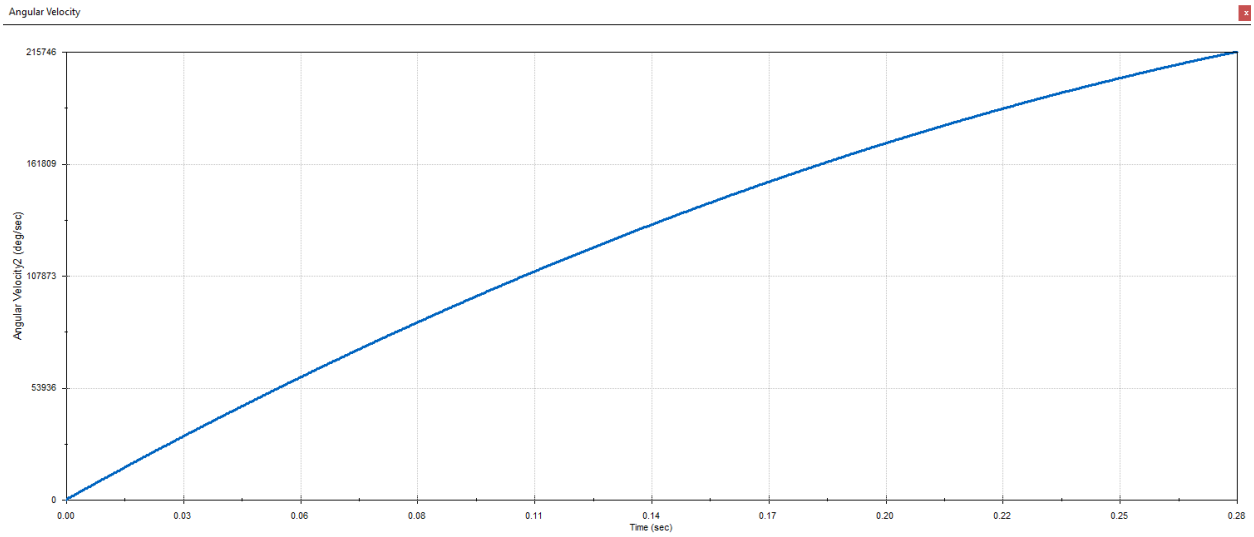


Figure 45: Angular Velocity of Propeller's Blade.

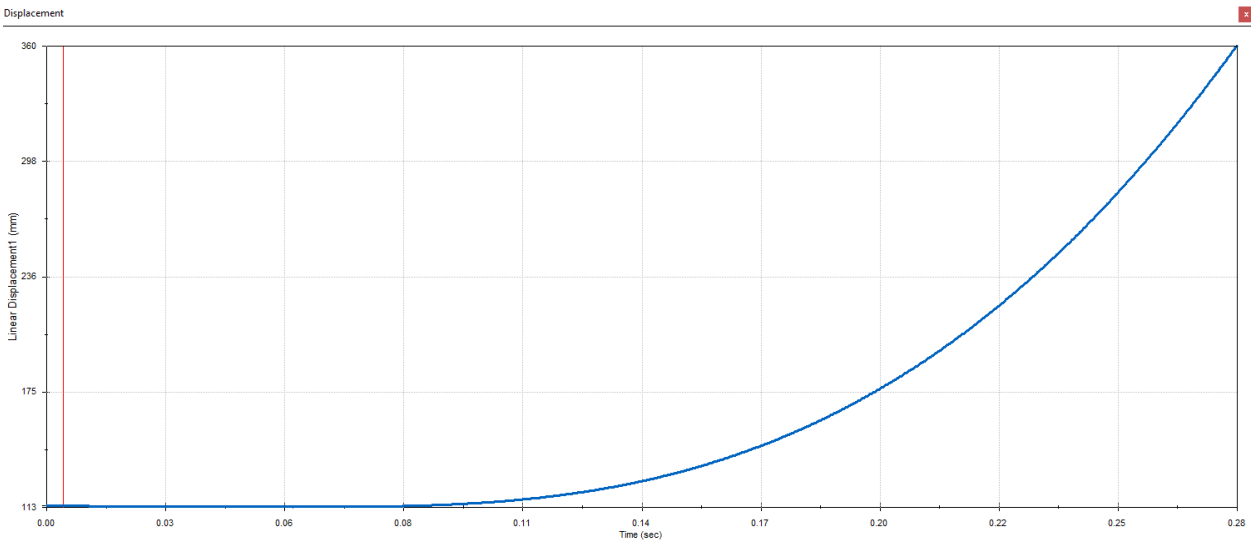


Figure 46: Linear Y Displacement of The Main Frame.

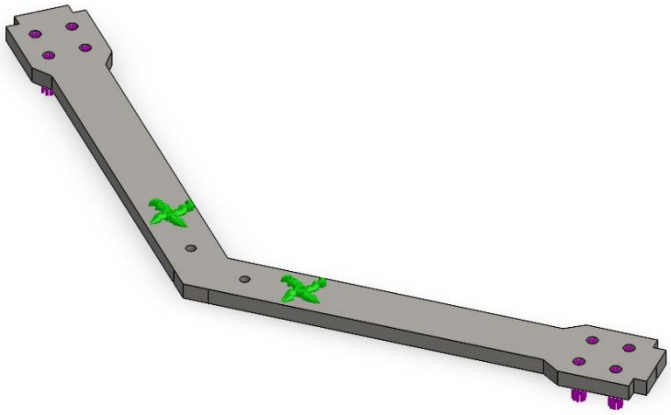
We applied angular velocity on the propellers and a thrust force on the body of the drone to simulate it flying to a cruise height by applying values as data points to simulate it taking off from ground, then gradually reaching a cruising height. (Note we used 4000 FPS).

Stress Simulation:

Assumptions

We applied fixtures in place where the screws are placed, then we calculated the thrust force of each propeller used previously (5 N) on each side of the bracket since each V bracket is mounted with two motors on opposite ends.

Model Information



Model name: V bracket (V)
Current Configuration: Default

Solid Bodies

Document Name and Reference	Treated As	Volumetric Properties	Document Path/Date Modified
Cut-Extrude2 	Solid Body	Mass:0.191255 kg Volume:2.45198e-05 m^3 Density:7,800 kg/m^3 Weight:1.87429 N	C:\Users\TD\Desktop\Final CAD Project\Our Project\Our Project\Saeed Parts\V bracket (V).SLDPRT Dec 11 23:39:13 2024

Study Properties

Study name	Stress Analysis
Analysis type	Static
Mesh type	Solid Mesh
Thermal Effect:	On
Thermal option	Include temperature loads
Zero strain temperature	298 Kelvin
Include fluid pressure effects from SOLIDWORKS Flow Simulation	Off
Solver type	Automatic
Inplane Effect:	Off
Soft Spring:	Off
Inertial Relief:	Off
Incompatible bonding options	Automatic
Large displacement	Off
Compute free body forces	On
Friction	Off
Use Adaptive Method:	Off
Result folder	SOLIDWORKS document (C:\Users\TD\Desktop\Final CAD Project\Our Project\Our Project\Saeed Parts)

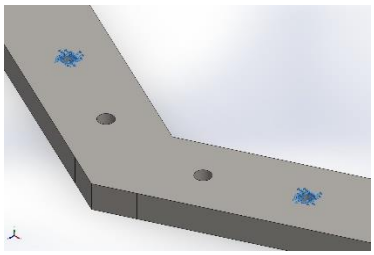
Units

Unit system:	SI (MKS)
Length/Displacement	mm
Temperature	Kelvin
Angular velocity	Rad/sec
Pressure/Stress	N/m^2

Material Properties

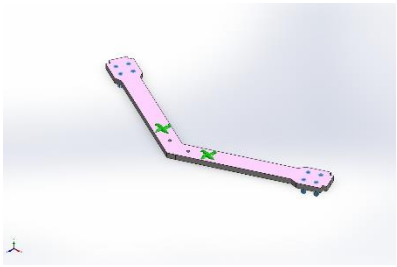
Model Reference	Properties		Components
	Name:	**Cast Carbon Steel**	**SolidBody 1(Cut-Extrude2)(V bracket (V))**
	Model type:	**Linear Elastic Isotropic**	
	Default failure criterion:	**Max von Mises Stress**	
	Yield strength:	**2.48168e+08 N/m^2**	
	Tensile strength:	**4.82549e+08 N/m^2**	
	Elastic modulus:	**2e+11 N/m^2**	
	Poisson's ratio:	**0.32**	
	Mass density:	**7,800 kg/m^3**	
	Shear modulus:	**7.6e+10 N/m^2**	
	Thermal expansion coefficient:	**1.2e-05 /Kelvin**	
Curve Data:N/A			

Loads and Fixtures

Fixture name	Fixture Image	Fixture Details
Fixed-1		**Entities:** 2 edge(s) **Type:** Fixed Geometry

Resultant Forces

Components	X	Y	Z	Resultant
Reaction force(N)	1.19209e-06	-6	3.33786e-06	6
Reaction Moment(N.m)	0	0	0	0

Load name	Load Image	Load Details
Force-1		**Entities:** 8 edge(s) **Reference:** Face< 1 > **Type:** Apply force **Values:** ---, ---, 0.75 N

Connector Definitions

No Data

Interaction Information

No Data

Mesh information

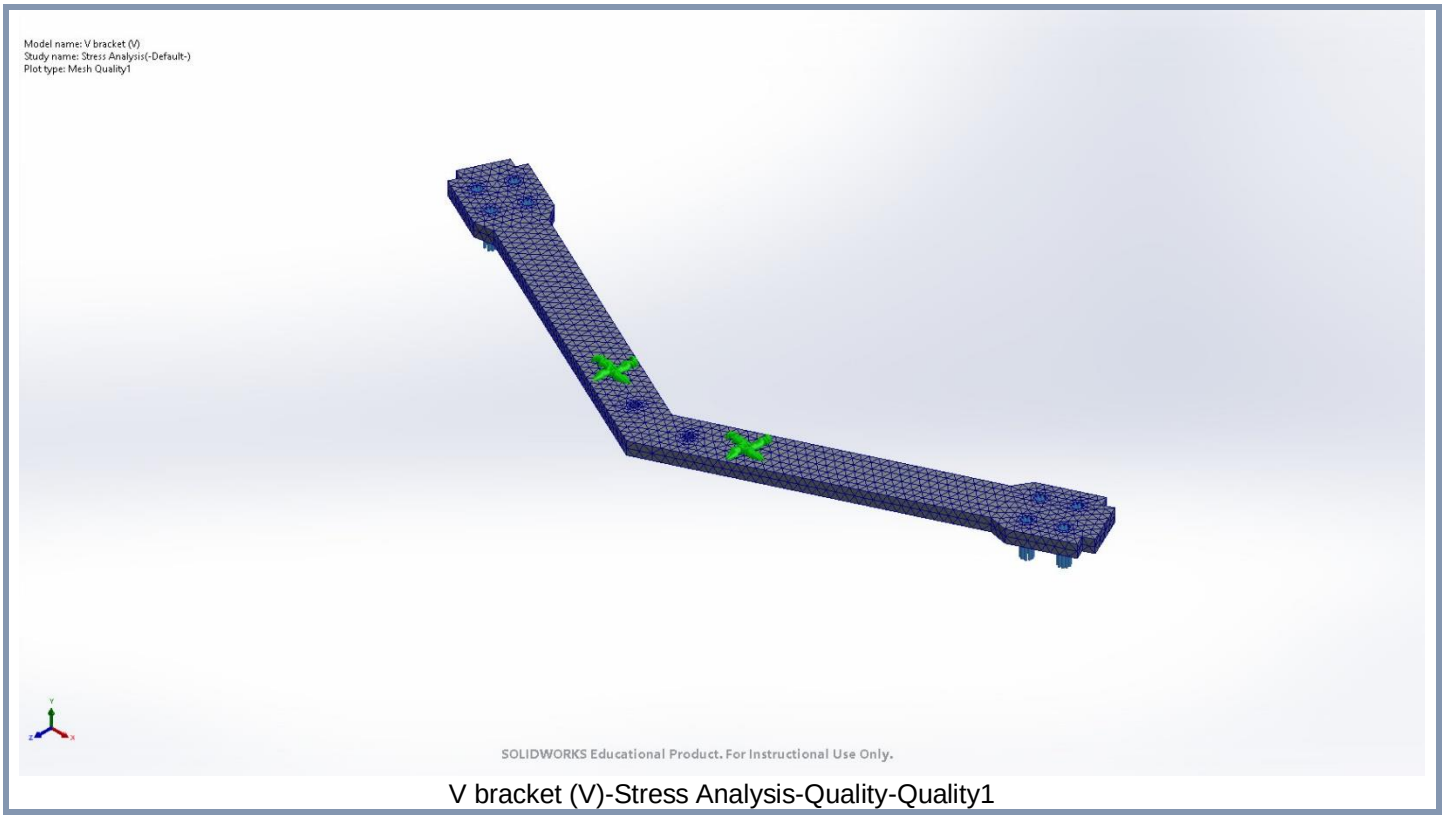
Mesh type	Solid Mesh
Mesher Used:	Blended curvature-based mesh
Jacobian points for High quality mesh	16 Points
Maximum element size	2.9062 mm
Minimum element size	0.968725 mm
Mesh Quality	High

Mesh information - Details

Total Nodes	21268
Total Elements	12113
Maximum Aspect Ratio	7.5588
% of elements with Aspect Ratio < 3	97.7
Percentage of elements with Aspect Ratio > 10	0
Percentage of distorted elements	0
Time to complete mesh(hh;mm;ss):	00:00:04
Computer name:	

Mesh Quality Plots

Name	Type	Min	Max
Quality1	Mesh	-	-



V bracket (V)-Stress Analysis-Quality-Quality1

Sensor Details

No Data

Resultant Forces

Reaction forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	1.19209e-06	-6	3.33786e-06	6

Reaction Moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	0

Free body forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	1.19209e-06	1.21593e-05	1.43051e-06	1.23011e-05

Free body moments

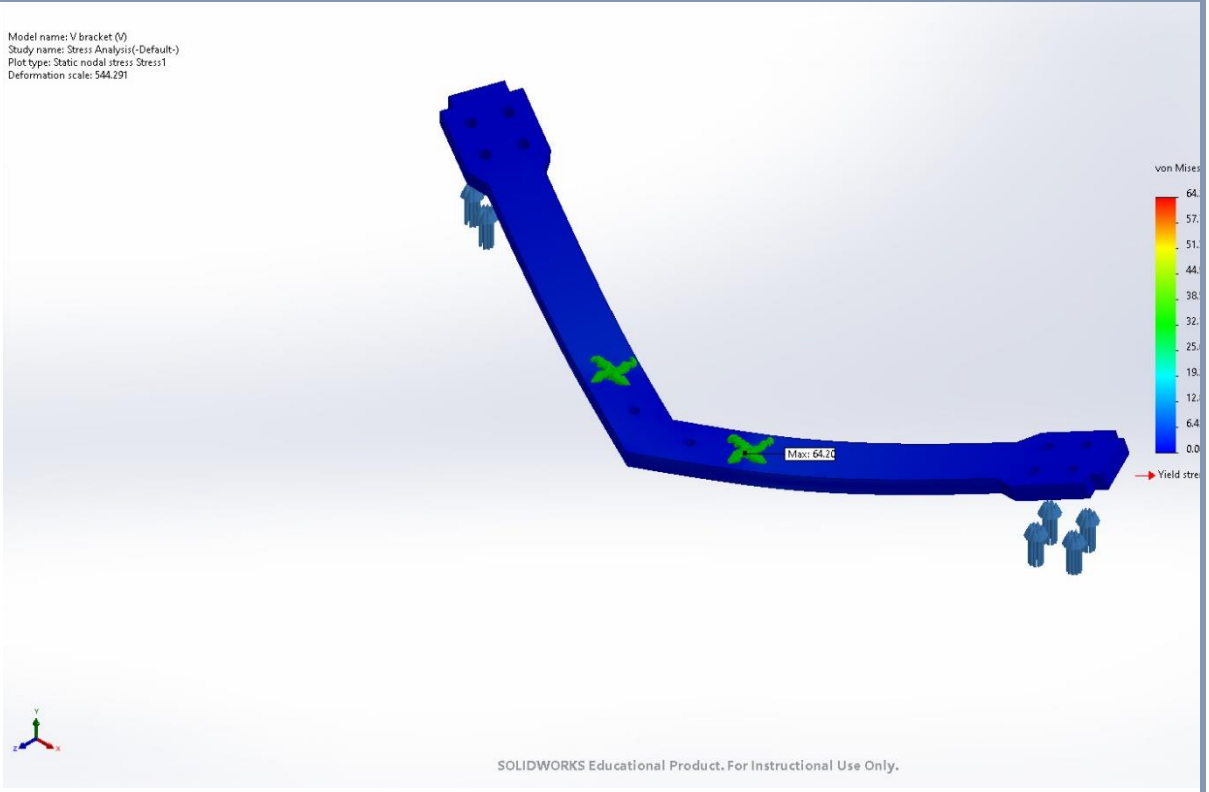
Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	1e-33

Beams

No Data

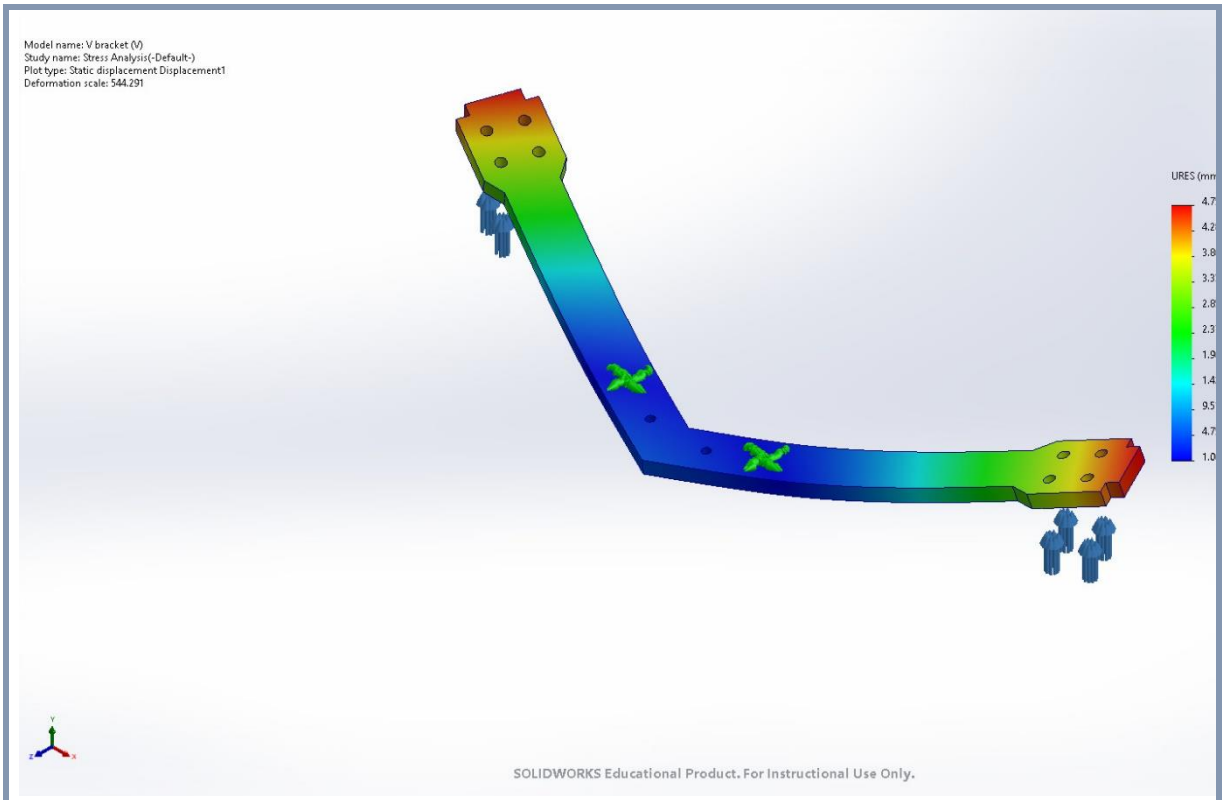
Study Results

Name	Type	Min	Max
Stress1	VON: von Mises Stress	0.00N/mm^2 (MPa) Node: 16535	64.20N/mm^2 (MPa) Node: 20145



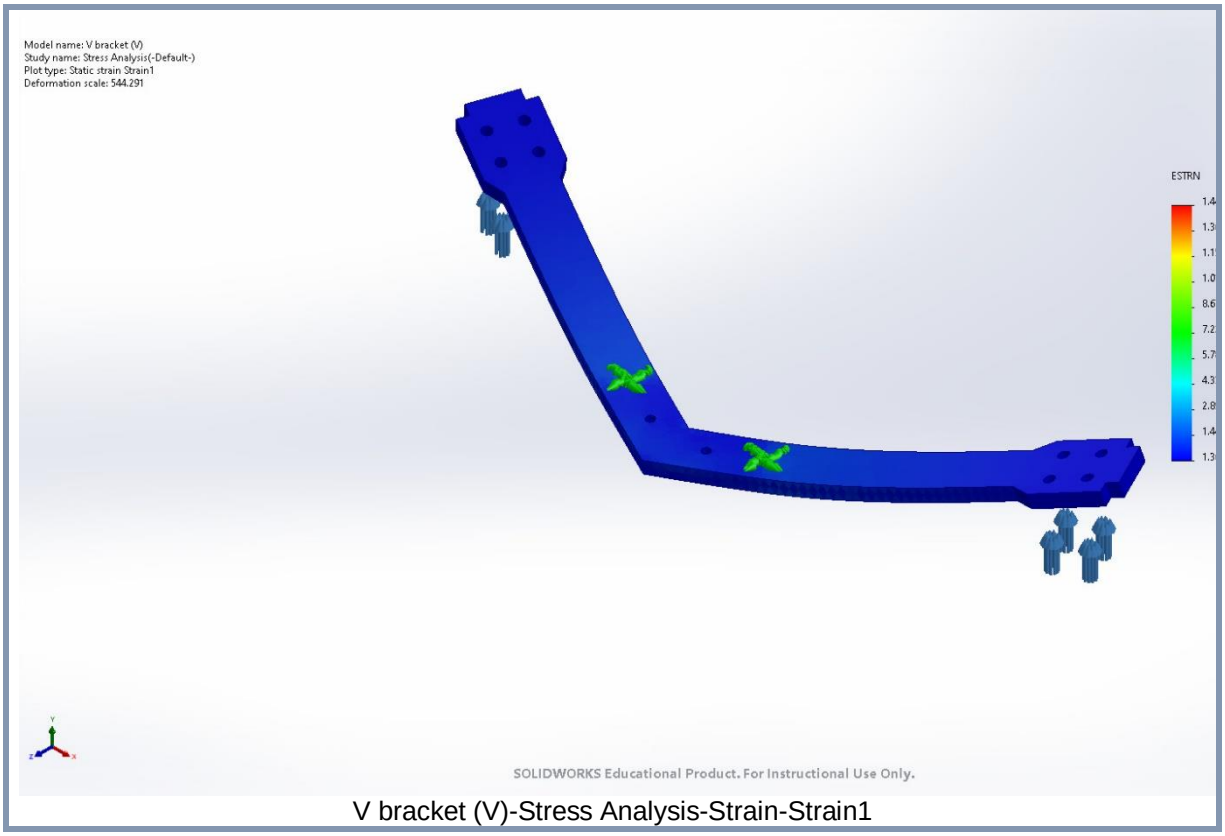
V bracket (V)-Stress Analysis-Stress-Stress1

Name	Type	Min	Max
Displacement1	URES: Resultant Displacement	0.000e+00mm Node: 69	4.758e-02mm Node: 17



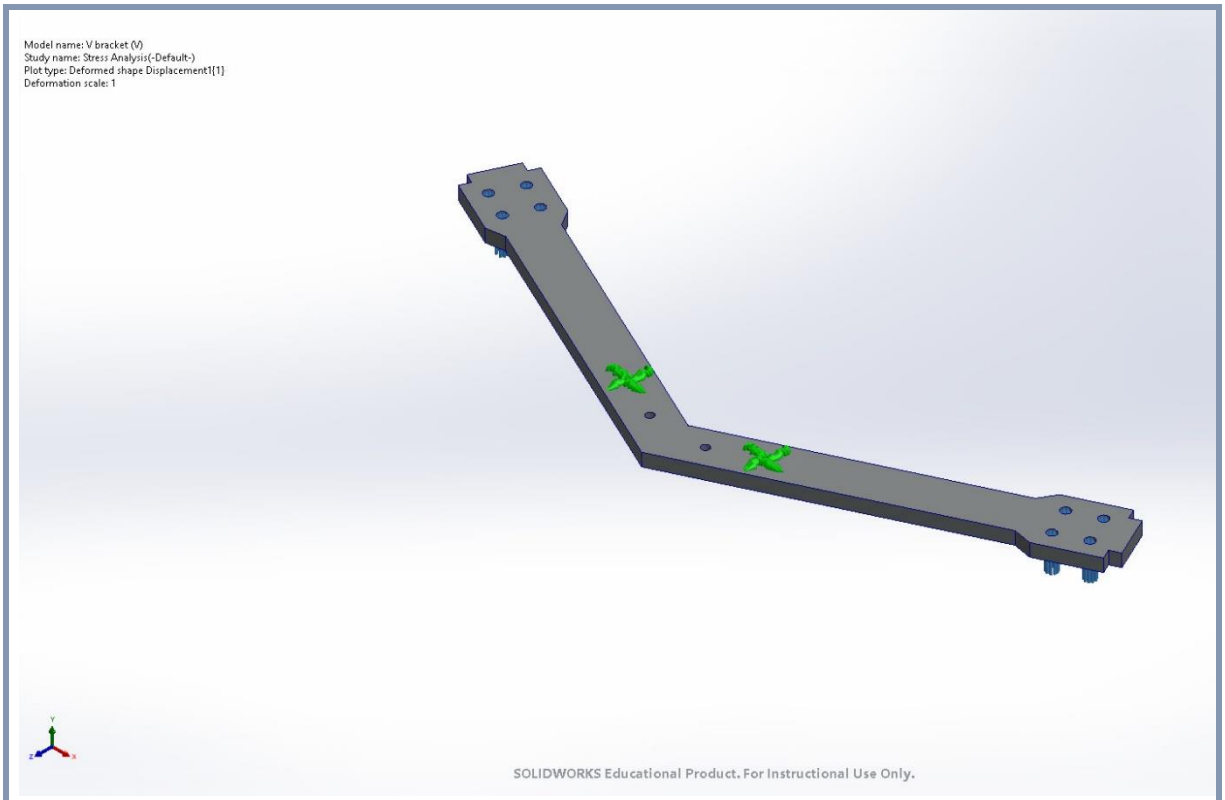
V bracket (V)-Stress Analysis-Displacement-Displacement1

Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain	1.307e-08 Element: 9410	1.445e-04 Element: 8387



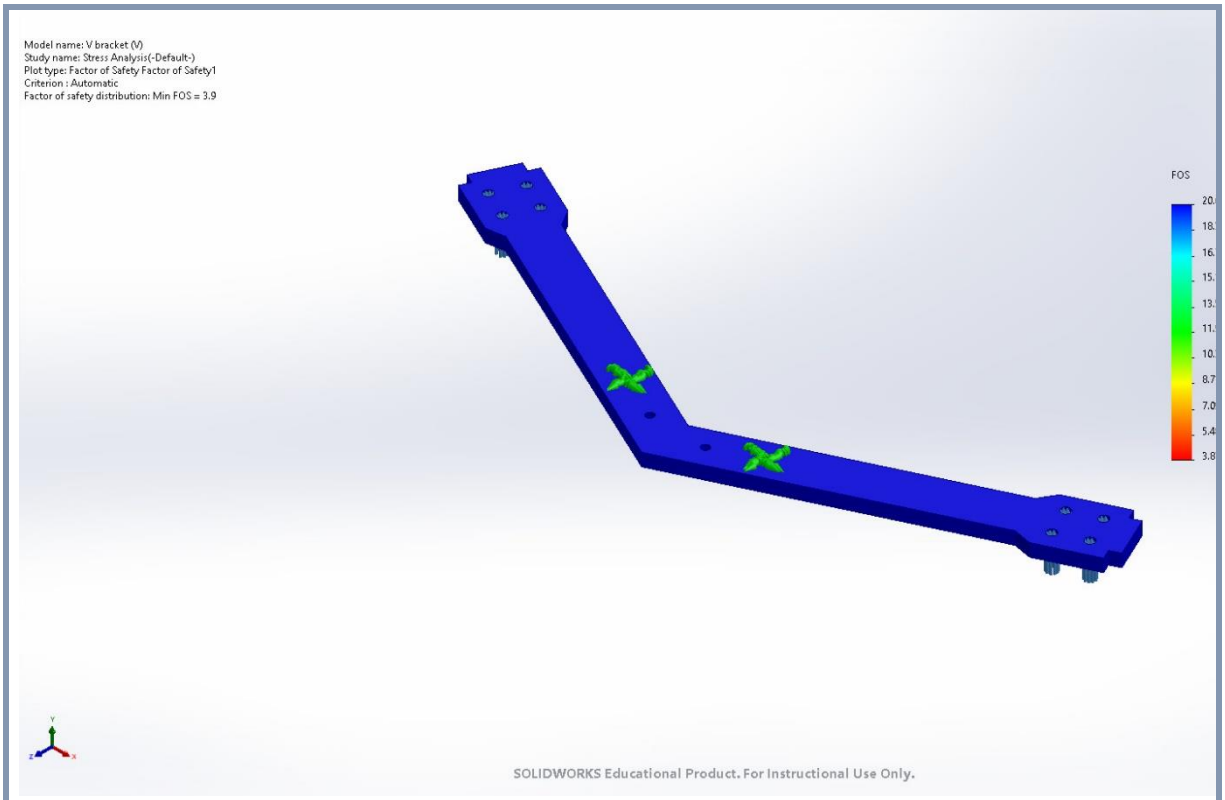
V bracket (V)-Stress Analysis-Strain-Strain1

Name	Type
Displacement1{1}	Deformed shape



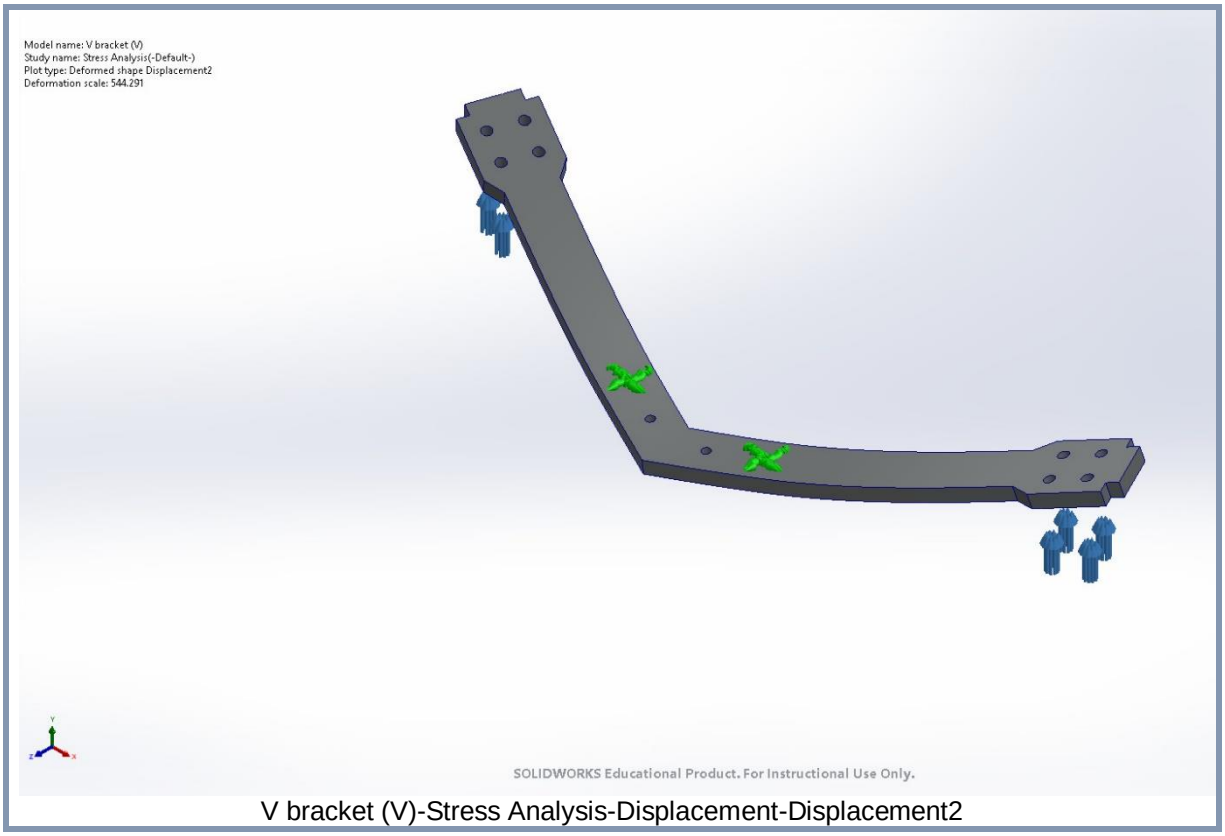
V bracket (V)-Stress Analysis-Displacement-Displacement1{1}

Name	Type	Min	Max
Factor of Safety1	Automatic	3.87 Node: 20145	1,693,870.63 Node: 16535



V bracket (V)-Stress Analysis-Factor of Safety-Factor of Safety1

Name	Type
Displacement2	Deformed shape



V bracket (V)-Stress Analysis-Displacement-Displacement2

Flow Analysis:

We faced a problem in defining the fluid volume every time we define one an error pops up saying zero thickness geometry, so we were not able to apply flow simulation.

Sustainability Analysis:

No Company

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Sustainability Report



Model Name:	Assembly Drone
Weight:	1255.43 g
Built to last:	1.0 year
Duration of use:	1.0 year



Manufacturing Region

The choice of manufacturing region determines the energy sources and technologies used in the modeled material creation and manufacturing steps of the product's life cycle.

Use Region

The use region is used to determine the energy sources consumed during the product's use phase (if applicable) and the destination for the product at its end-of-life. Together with the manufacturing region, the use region is also used to estimate the environmental impacts associated with transporting the product from its manufacturing location to its use location.

Summary

Learn more about Life Cycle Assessment

Sustainability Report

Model Name:	Assembly Drone	Weight:	1255.43 g
		Built to last:	1.0 year
		Duration of use:	1.0 year

Assembly Process

Region:	Asia
Energy type:	None
Energy amount:	0.00 kWh
Built to last:	1.0 year

Use

Region:	Asia
Energy type:	None
Energy amount:	0.00 kWh
Duration of use:	1.0 year

Transportation

Truck distance:	1600 km
Train distance:	0.00 km
Ship distance:	0.00 km
Airplane Distance:	0.00 km

End of Life

Recycled:	33 %
Incinerated:	13 %
Landfill:	54 %

Comments

Click here for alternative units such as 'Miles Driven in a Car'

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Sustainability Report

Model Name:	Assembly Drone	Weight:	1255.43 g
		Built to last:	1.0 year
		Duration of use:	1.0 year

Environmental Impact (calculated using CML impact assessment methodology)

Carbon Footprint



Material:	18 kg CO_2e
Manufacturing:	550 kg CO_2e
Use:	0.00 kg CO_2e
Transportation:	0.107 kg CO_2e
End of Life:	0.690 kg CO_2e

570 kg CO_2e

Total Energy Consumed



Material:	260 MJ
Manufacturing:	5500 MJ
Use:	0.00 MJ
Transportation:	1.4 MJ
End of Life:	0.507 MJ

5800 MJ

Air Acidification



Material:	0.092 kg SO_2e
Manufacturing:	7.8 kg SO_2e
Use:	0.00 kg SO_2e
Transportation:	5.4E-4 kg SO_2e
End of Life:	3.5E-4 kg SO_2e

7.9 kg SO_2e

Water Eutrophication



Material:	5.2E-3 kg PO_4e
Manufacturing:	0.302 kg PO_4e
Use:	0.00 kg PO_4e
Transportation:	1.1E-4 kg PO_4e
End of Life:	8.7E-4 kg PO_4e

0.308 kg PO_4e

Material Financial Impact 2.20 USD

Comments

Click here for alternative units such as 'Miles Driven in a Car'

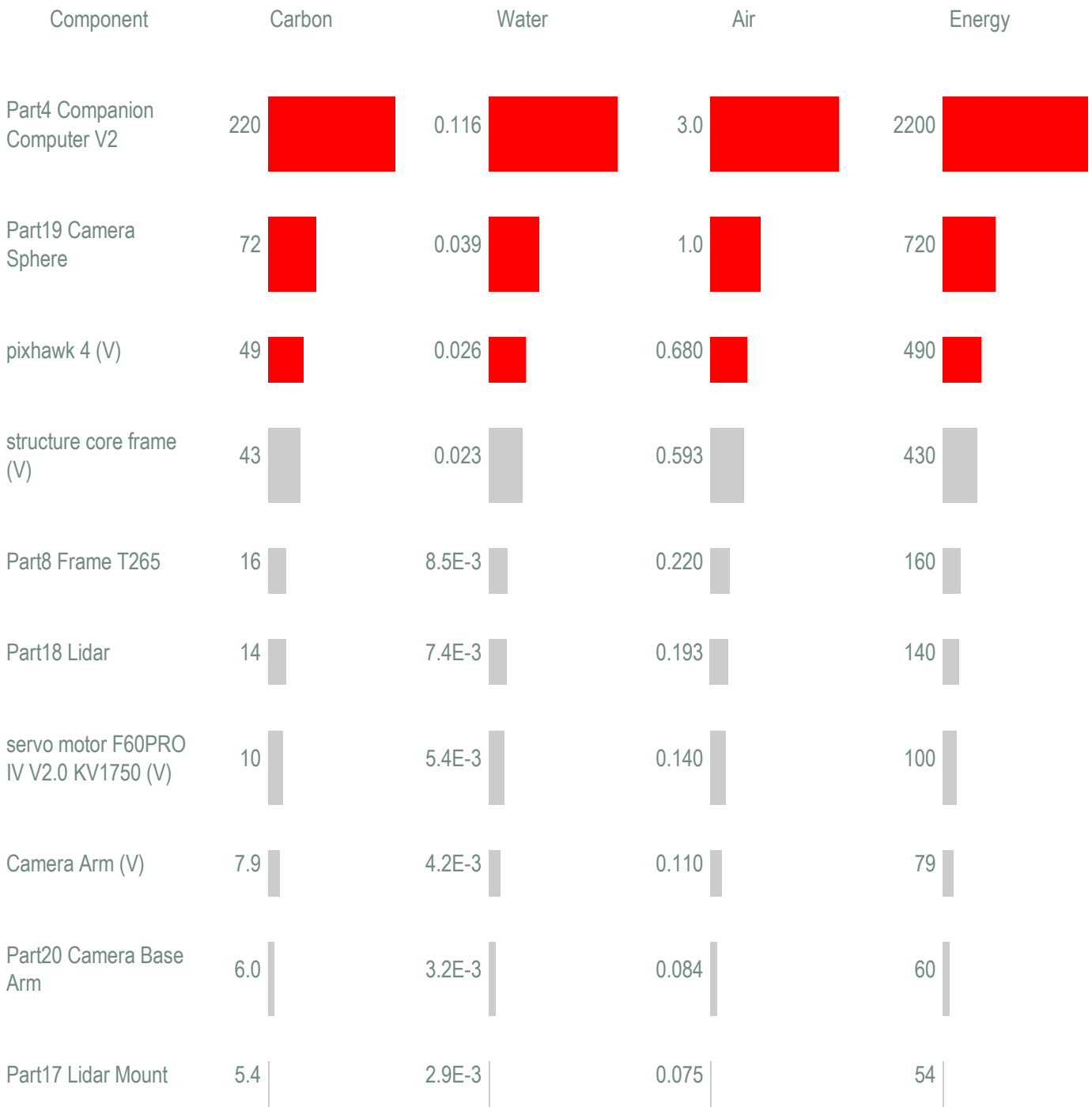
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Sustainability Report

Model Name:	Assembly Drone	Weight:	1255.43 g
		Built to last:	1.0 year
		Duration of use:	1.0 year

Component Environmental Impact

Top Ten Components Contributing Most to the Four Areas of Environmental Impact


Component
Carbon
Water
Air
Energy
Part4 Companion Computer V2
220
0.116
3.0
2200
Part19 Camera Sphere
72
0.039
1.0
720
pixhawk 4 (V)
49
0.026
0.680
490
structure core frame (V)
43
0.023
0.593
430
Part8 Frame T265
16
8.5E-3
0.220
160
Part18 Lidar
14
7.4E-3
0.193
140
servo motor F60PRO IV V2.0 KV1750 (V)
10
5.4E-3
0.140
100
Camera Arm (V)
7.9
4.2E-3
0.110
79
Part20 Camera Base Arm
6.0
3.2E-3
0.084
60
Part17 Lidar Mount
5.4
2.9E-3
0.075
54



Comments

[Click here for alternative units such as ‘Miles Driven in a Car’]

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Sustainability Report


RAW MATERIAL
EXTRACTION
MATERIAL
PROCESSING
PART
MANUFACTURING
ASSEMBLY
PRODUCT
USE
END OF LIFE
SUSTAINABLE
DESIGN

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Sustainability Report

Glossary

Air Acidification - Sulfur dioxide, nitrous oxides other acidic emissions to air cause an increase in the acidity of rainwater, which in turn acidifies lakes and soil. These acids can make the land and water toxic for plants and aquatic life. Acid rain can also slowly dissolve manmade building materials such as concrete. This impact is typically measured in units of either kg **sulfur dioxide equivalent (SO_2), or moles H+ equivalent**.

Carbon Footprint - Carbon-dioxide and other gasses which result from the burning of fossil fuels accumulate in the atmosphere which in turn increases the earth's average temperature. Carbon footprint acts as a proxy for the larger impact factor referred to as Global Warming Potential (GWP). Global warming is blamed for problems like loss of glaciers, extinction of species, and more extreme weather, among others.

Total Energy Consumed - A measure of the non-renewable energy sources associated with the part's lifecycle in units of megajoules (**MJ**). This impact includes not only the electricity or fuels used during the product's lifecycle, but also the upstream energy required to obtain and process these fuels, and the embodied energy of materials which would be released if burned. PED is expressed as the net calorific value of energy demand from non-renewable resources (e.g. petroleum, natural gas, etc.). Efficiencies in energy conversion (e.g. power, heat, steam, etc.) are taken into account.

Water Eutrophication - When an over abundance of nutrients are added to a water ecosystem, eutrophication occurs. Nitrogen and phosphorous from waste water and agricultural fertilizers causes an overabundance of algae to bloom, which then depletes the water of oxygen and results in the death of both plant and animal life. This impact is typically measured in either kg **phosphate equivalent (PO_4) or kg nitrogen (N) equivalent**.

Life Cycle Assessment (LCA) - This is a method to quantitatively assess the environmental impact of a product throughout its entire lifecycle, from the procurement of the raw materials, through the production, distribution, use, disposal and recycling of that product.

Material Financial Impact - This is the financial impact associated with the material only. The mass of the model is multiplied by the financial impact unit (units of currency/units of mass) to calculate the financial impact (in units of currency).

Learn more about Life Cycle Assessment

References:

Drone: https://grabcad.com/library/drone-race-px-1

Camera: https://grabcad.com/library/dji-style-gimbal-with-camera-1

Note that for the camera we modified the dimensions to satisfy our needs.